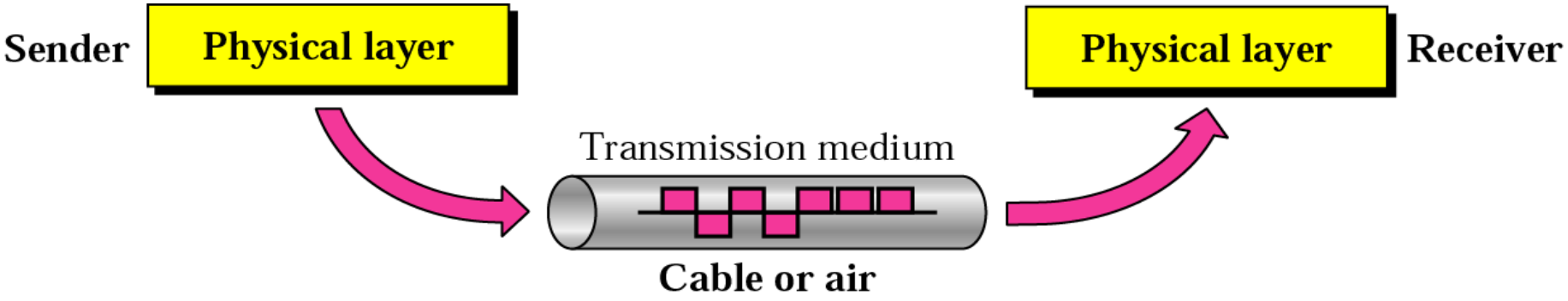
Data Communications System

Compiled By

Prabhakar Singh

Chapter 5 – Transmission Media



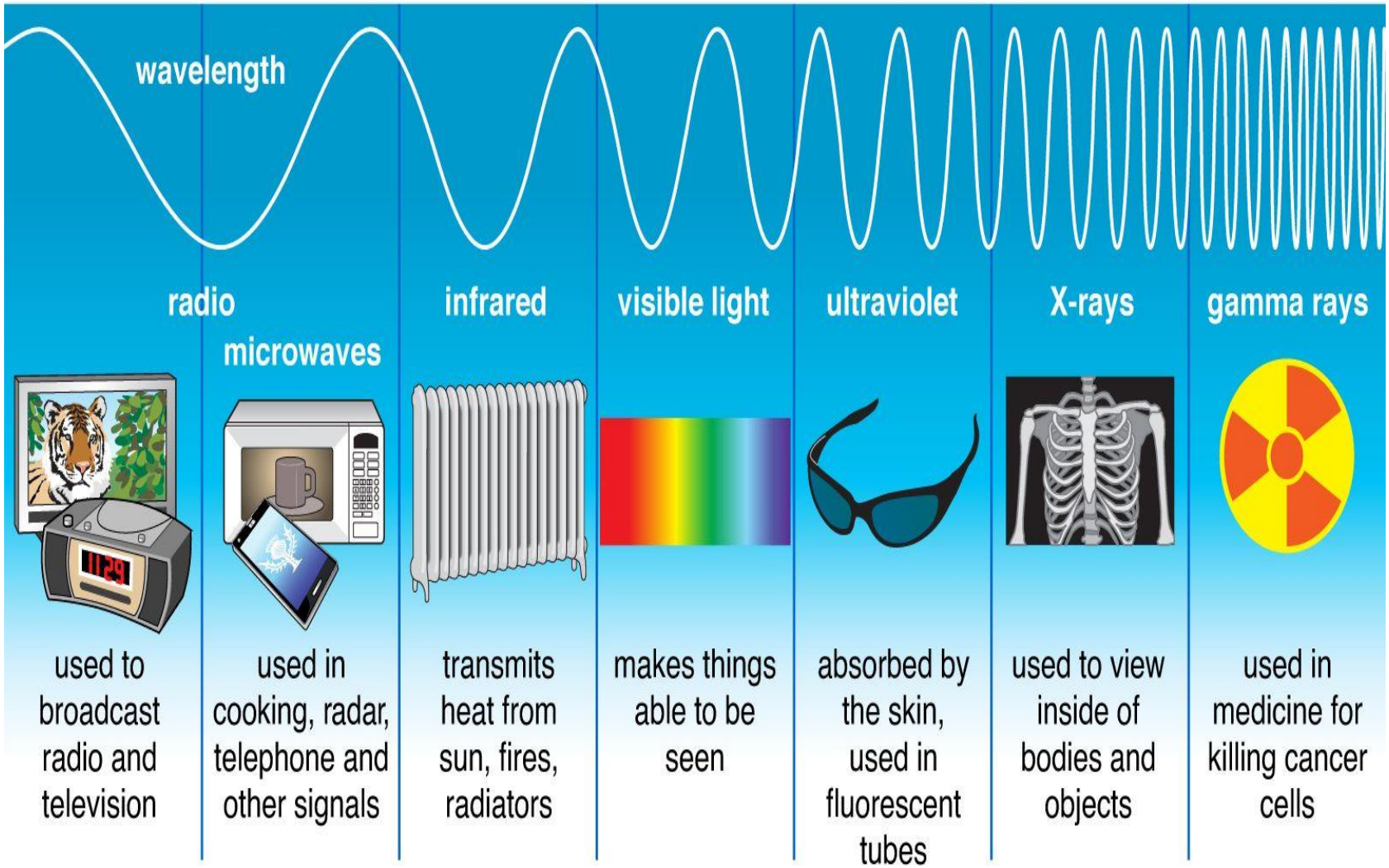
Transmission Media

Introduction

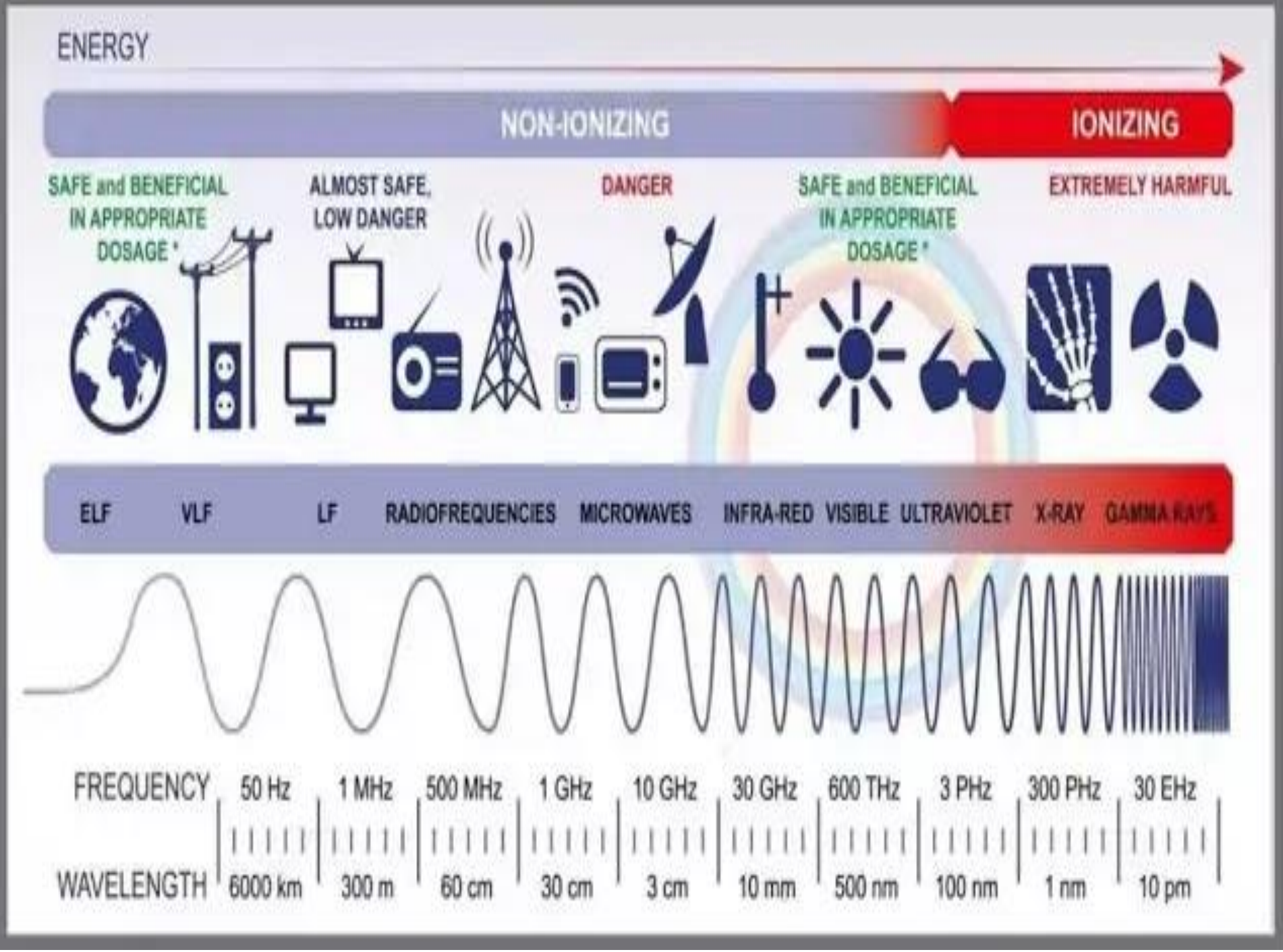
- Transmission medium may be anything that can carry information from a source to a destination .E.g. Conversation between two person have air as transmission medium.
- Signals can travel through a vacuum, air, or other media
- In data communication ,it works like a **physical path between the sender and the receiver** . For E.g. in a copper cable network the bits in the form of electrical signals whereas in a fiber network ,the bits are available in the form of light pulses.
- For long distance transmission ,the data are converted in **EM signals** .EM is the combination of electric and magnetic field where both are perpendicular to each other, it includes radio waves, infrared light, visible lights , UV lights , cosmic lights etc.

Band	Range	Propagation	Application
VLF(Very Low Frequency)	**3-30KHz**	**Ground**	Long range radio navigation
LF (Low Frequency)	**30-300KHz**	**Ground**	Radio beacons & navigation locator
MF(Medium Frequency)	**300KHz-3MHz**	**Sky**	AM radio
HF (High Frequency)	**3-30MHz**	**Sky**	Shortwave transmission , aircraft communication
VHF (Very High Frequency)	**30-300MHz**	**Sky & Line of Sight**	FM radio , TV broadcasting
UHF (Ultra High Frequency)	**300MHz-3GHz**	**Line of Sight**	Cellular phones, paging , satellite
SHF(Super High Frequency)	**3-30GHz**	**Line of Sight**	Satellite communication
EHF(Extremely High Frequency)	**30-300GHz**	**Line of Sight**	Radar , satellite

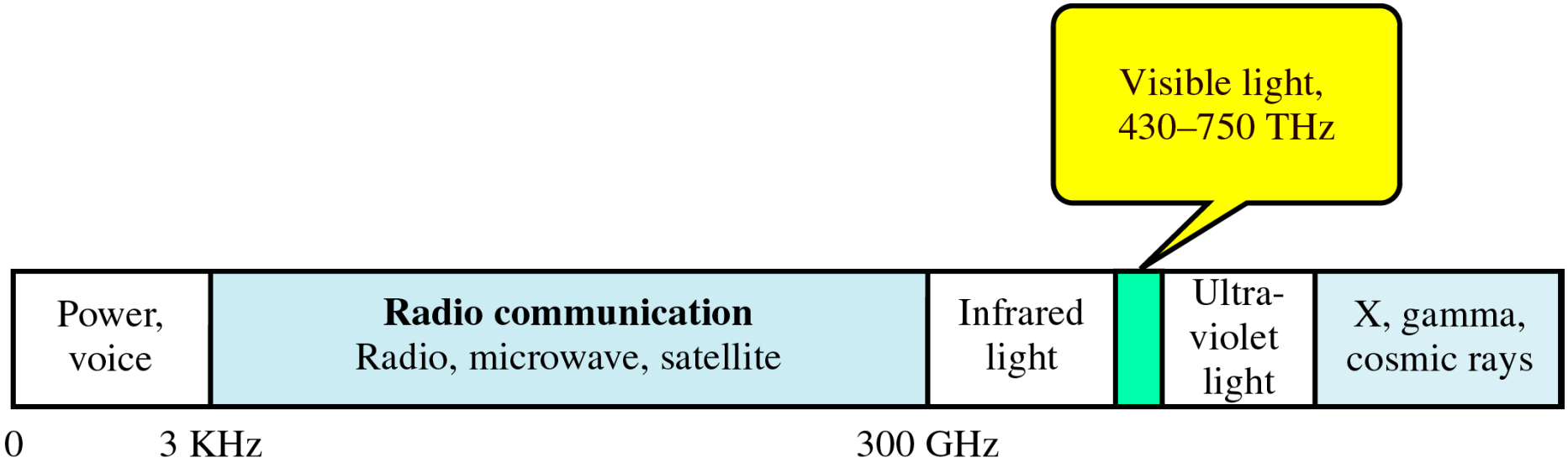
Types of Electromagnetic Radiation



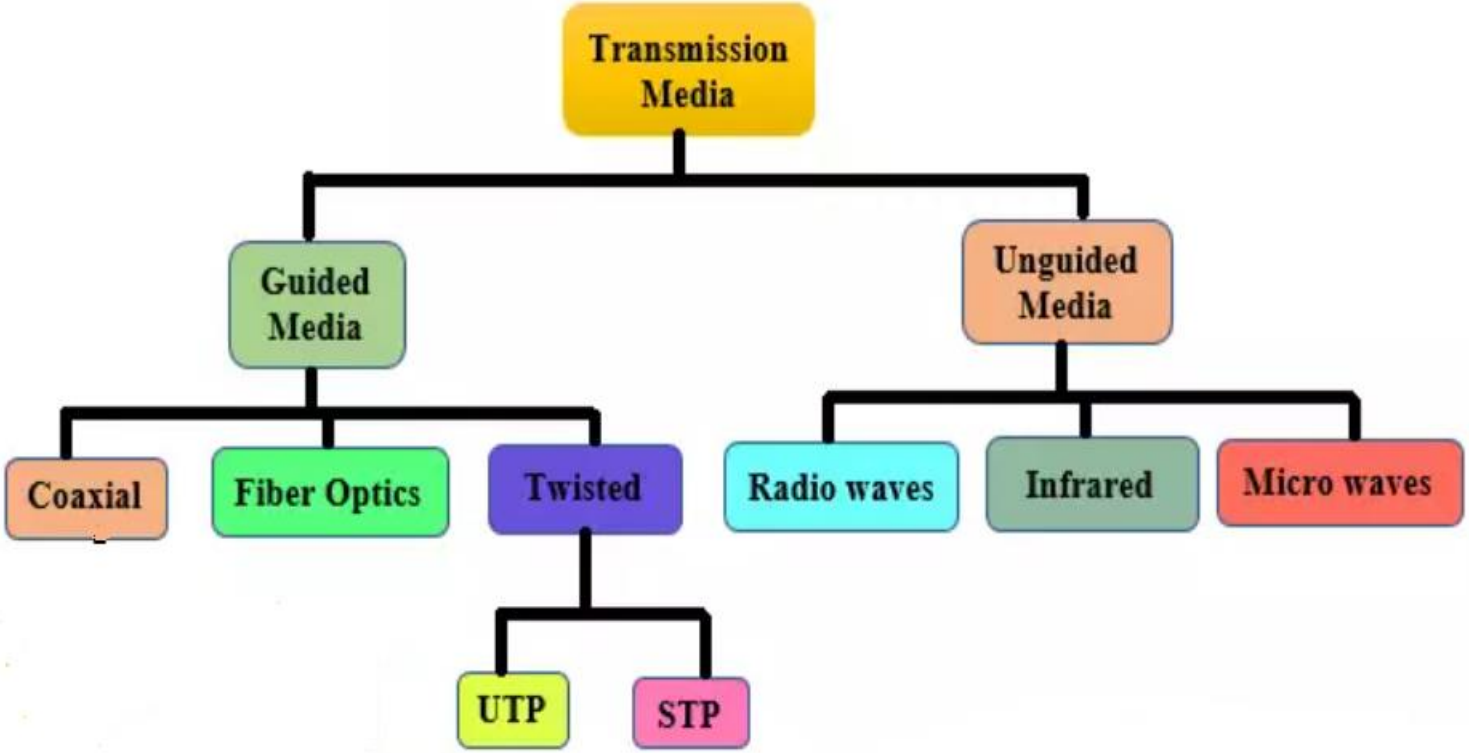
© Encyclopædia Britannica, Inc.


ENERGY
NON-IONIZING
IONIZING
SAFE and BENEFICIAL IN APPROPRIATE DOSAGE *
ALMOST SAFE, LOW DANGER
DANGER
SAFE and BENEFICIAL IN APPROPRIATE DOSAGE *
EXTREMELY HARMFUL
ELF
VLF
LF
RADIOFREQUENCIES
MICROWAVES
INFRA-RED
VISIBLE
ULTRAVIOLET
X-RAY
GAMMA RAYS
FREQUENCY
50 Hz
1 MHz
500 MHz
1 GHz
10 GHz
30 GHz
600 THz
3 PHz
300 PHz
30 EHz
WAVELENGTH
6000 km
300 m
60 cm
30 cm
3 cm
10 mm
500 nm
100 nm
1 nm
10 pm

Electromagnetic Spectrum



Types of Transmission Media



Categories of Media

- Guided Media
  - Twisted pair, coaxial, and fiber-optic
- Unguided Media
  - Radio waves, infrared light, visible light, and X, gamma, and cosmic rays
  - Sent by microwave, satellite, and cellular transmission

Guided Media

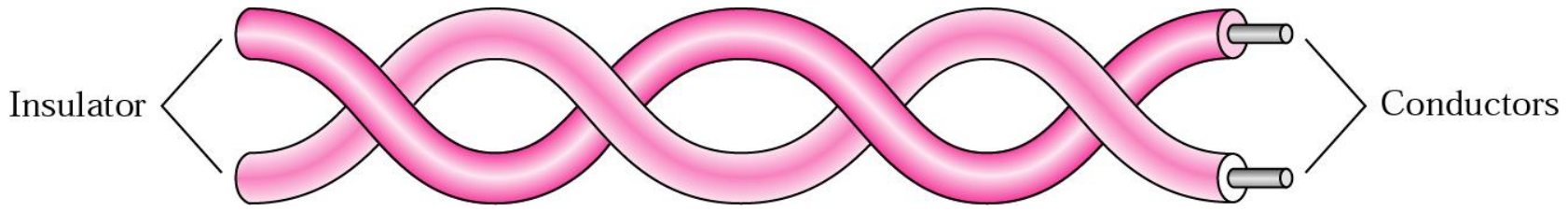
- Guided media is wired transmission media that guides the data signals along a specific path. Data signals are bound by the cabling system.
- Twisted-pair and coaxial use copper conductors to accept and transport signals in form of electrical current
- Optical fiber is glass or plastic cable that accepts and transports signals in form of light.

Unguided Media

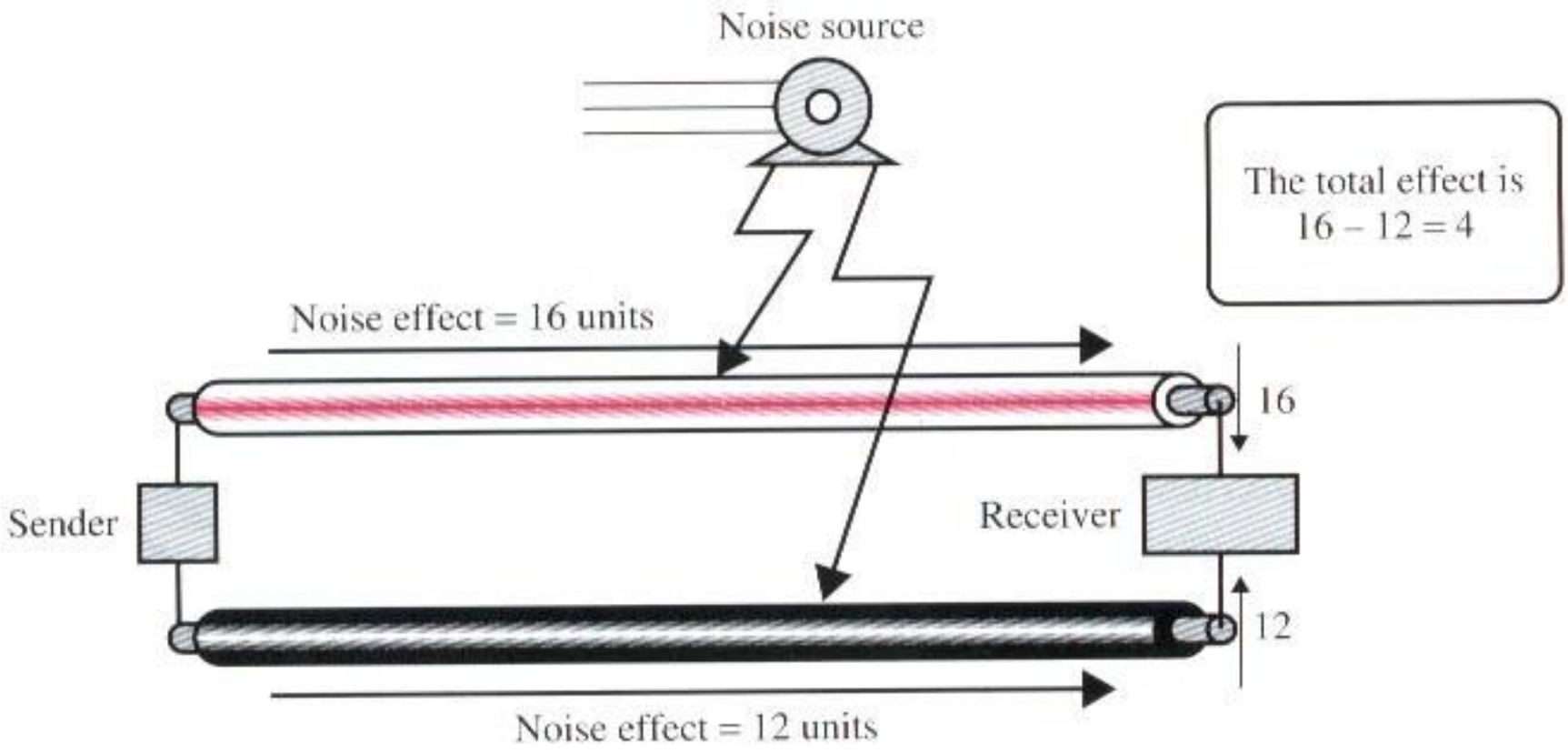
- Consists of a means for the data signals to travel but nothing to guide them along a specific path.
- For guided media , transmission & reception are achieved by means of an antenna.
- Wireless medium like air(Satellite ,Line-of-Sight ,VSAT) is medium.

Twisted-Pair Cable

- Two conductors surrounded by insulating material
- One wire used to carry signals, other used as a ground reference
- Twisting wires reduces the effect of noise interference or crosstalk since both wires will likely be equally affected
- Typically, a number of these pairs are bundled together
- For long distance communication hundreds of pairs of cables are used
- Least expensive & most widely used guided transmission medium.

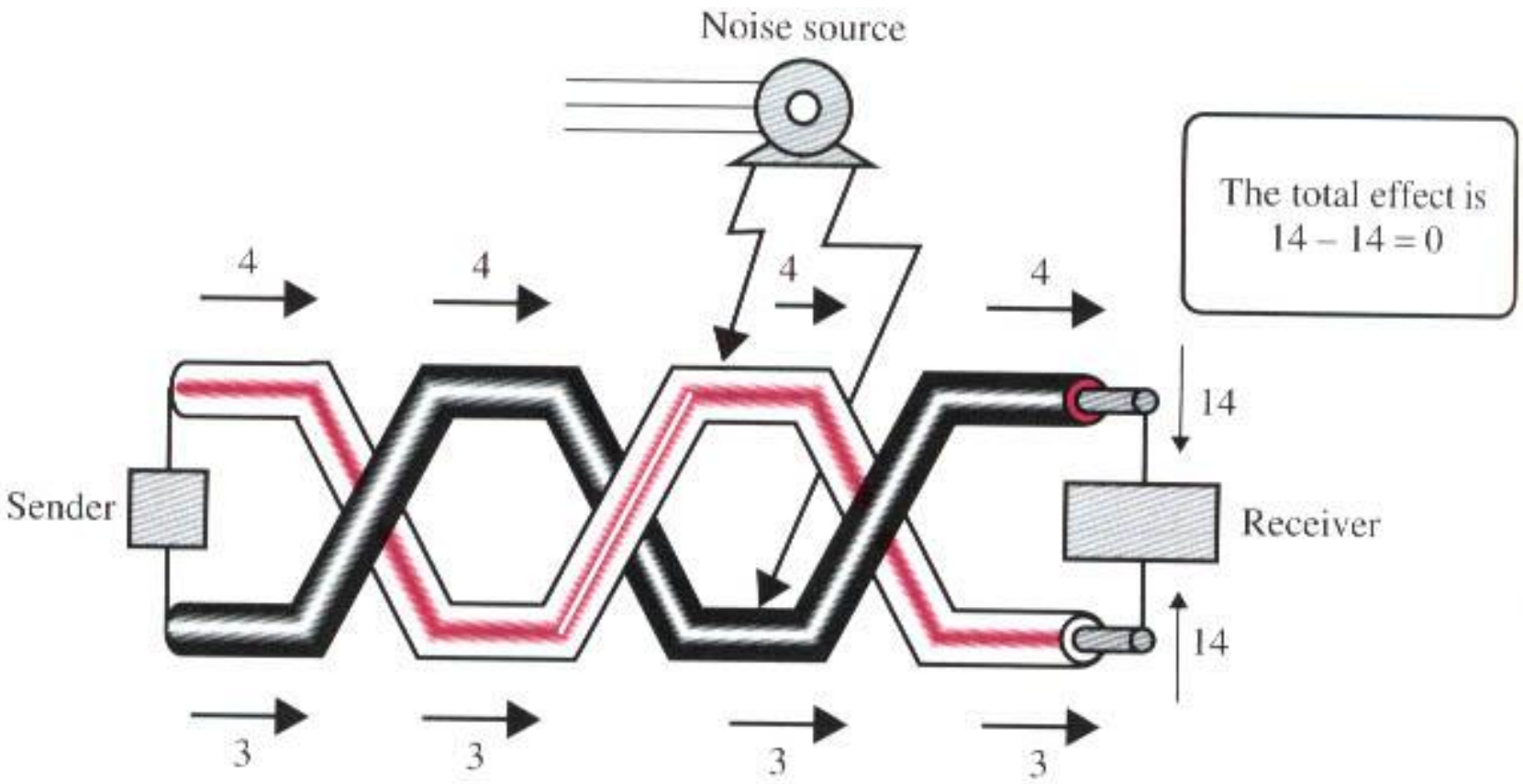


Effect of noise on parallel line

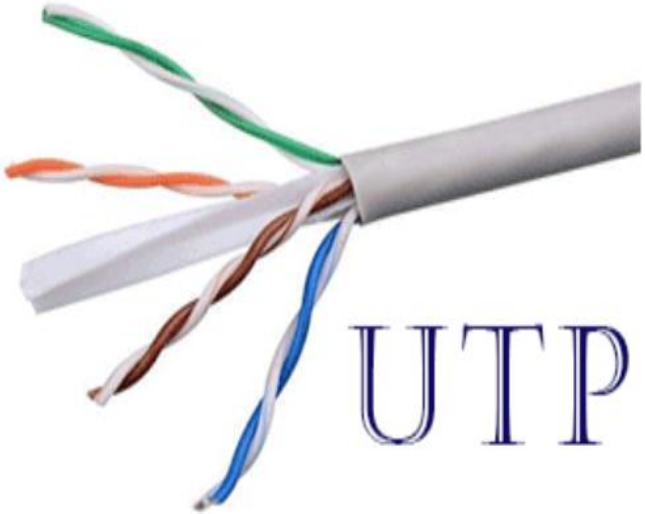


Effect of noise on parallel line

Effect of noise on twisted-pair lines



STP

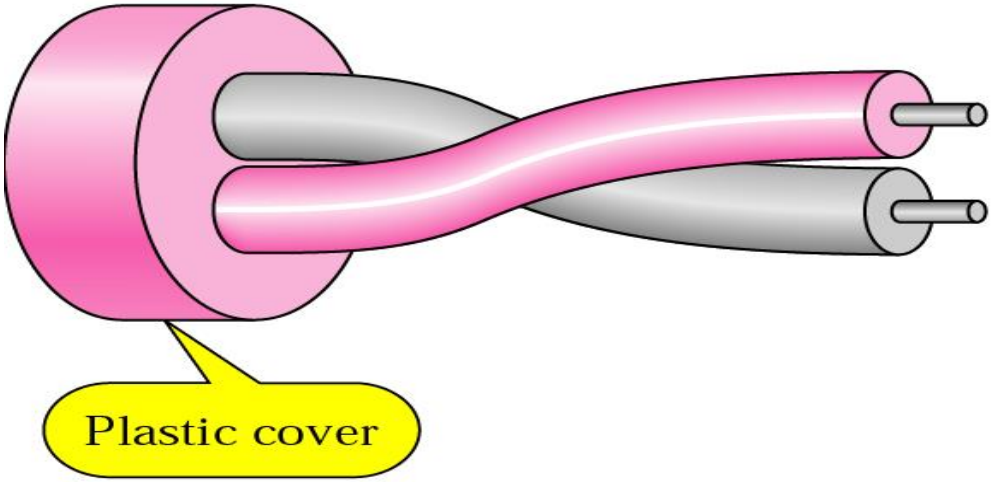

UTP

STP Cables

Unshielded Twisted Pair (UTP)

- Most common type, suitable for both voice and data transmission
- Typical impedance for UTP is 100 Ohm for Ethernet 10BaseT cable.
- Frequency range is from 100 Hz to 5 MHz
- Categories are determined by cable quality
  - Cat 3 commonly used for telephone systems (up to 10 Mbps)
  - Cat 5e usually used for data networks (up to 100 Mbps)
- Performance is measured by attenuation versus frequency and distance
- Adv: cheaper, flexible, easy to install
- UTP connectors attach to wire, most commonly used is RJ45, which have 8 conductors, one for each of four twisted pairs

UTP example

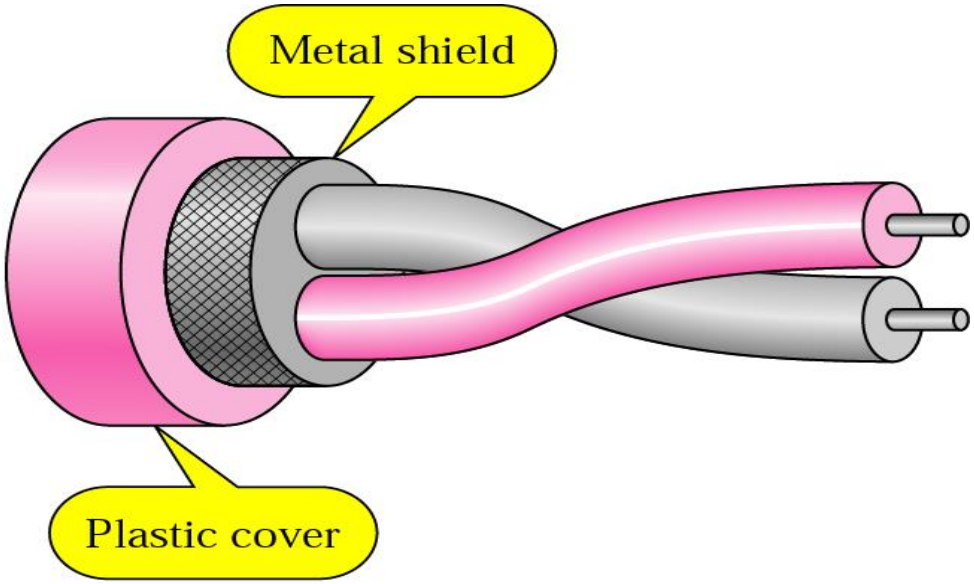


a. UTP

Shielded Twisted Pair (STP)

- IBM standardized STP cable types for their Token Ring protocol
- Provides better performance at higher data rates.
- A metal foil or braided-mesh covering encases each pair of insulated conductors to prevent electromagnetic noise called crosstalk
- Uses same connectors
- More expensive but less susceptible to noise

STP example



b. STP

Category	Max. Data Rate	Bandwidth	Max. Distance	Usage
Category 1	1 Mbps	0.4 MHz		Telephone and modem lines
Category 2	4 Mbps	4 MHz		LocalTalk & Telephone
Category 3	10 Mbps	16 MHz	100 m (328 ft.)	10BaseT Ethernet
Category 4	16 Mbps	20 MHz	100 m (328 ft.)	Token Ring
Category 5	100 Mbps	100 MHz	100 m (328 ft.)	100BaseT Ethernet
Category 5e	1 Gbps	100 MHz	100 m (328 ft.)	100BaseT Ethernet, residential homes
Category 6	1 Gbps	250 MHz	100 m (328 ft.) 10Gb at 37 m (121 ft.)	Gigabit Ethernet, commercial buildings
Category 6a	10 Gbps	500 MHz	100 m (328 ft.)	Gigabit Ethernet in data centers and commercial buildings
Category 7	10 Gbps	600 MHz	100 m (328 ft.)	10 Gbps Core Infrastructure
Category 7a	10 Gbps	1000 MHz	100 m (328 ft.) 40Gb at 50 m (164 ft.)	10 Gbps Core Infrastructure
Category 8	25 Gbps (Cat8.1) 40 Gbps (Cat8.2)	2000 MHz	30 m (98 ft.)	25 Gbps/40 Gbps Core Infrastructure

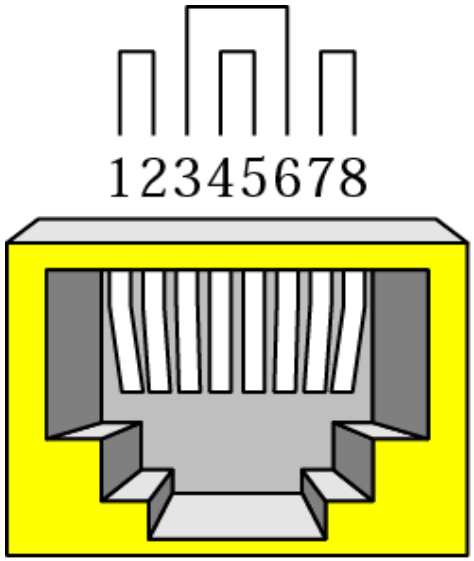
Categories of unshielded twisted-pair cables

Category	Max Speed	Digital/Analog	Use
1	< 100 kbps	Analog	Telephone
2	2 Mbps	Analog/digital	T-1 lines
3	10 Mbps	Digital	LANs
4	16 Mbps	Digital	LANs
5e	1 Gbps	Digital	LANs
6	1 Gbps	Digital	LANs
7 (Draft)	10 Gbps	Digital	LANs

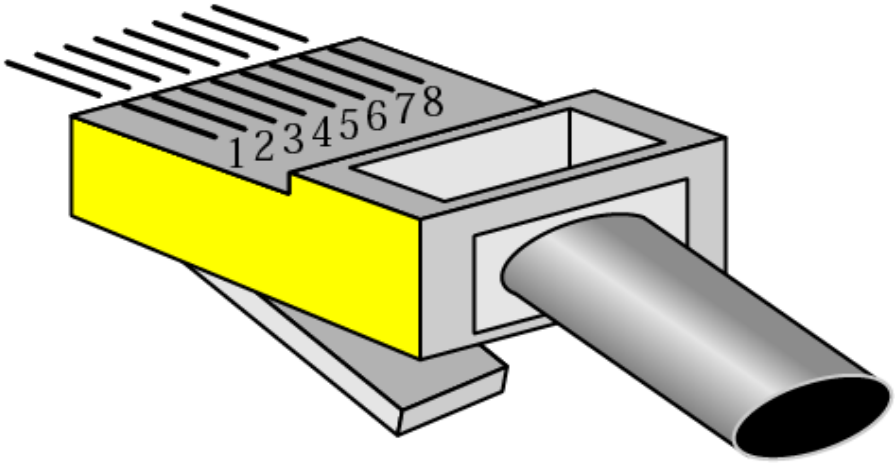
Color code for 4 cable pairs

Pair #	Primary color	Secondary color (stripe)
1	Blue	White
2	Orange	White
3	Green	White
4	Brown	White

UTP connector

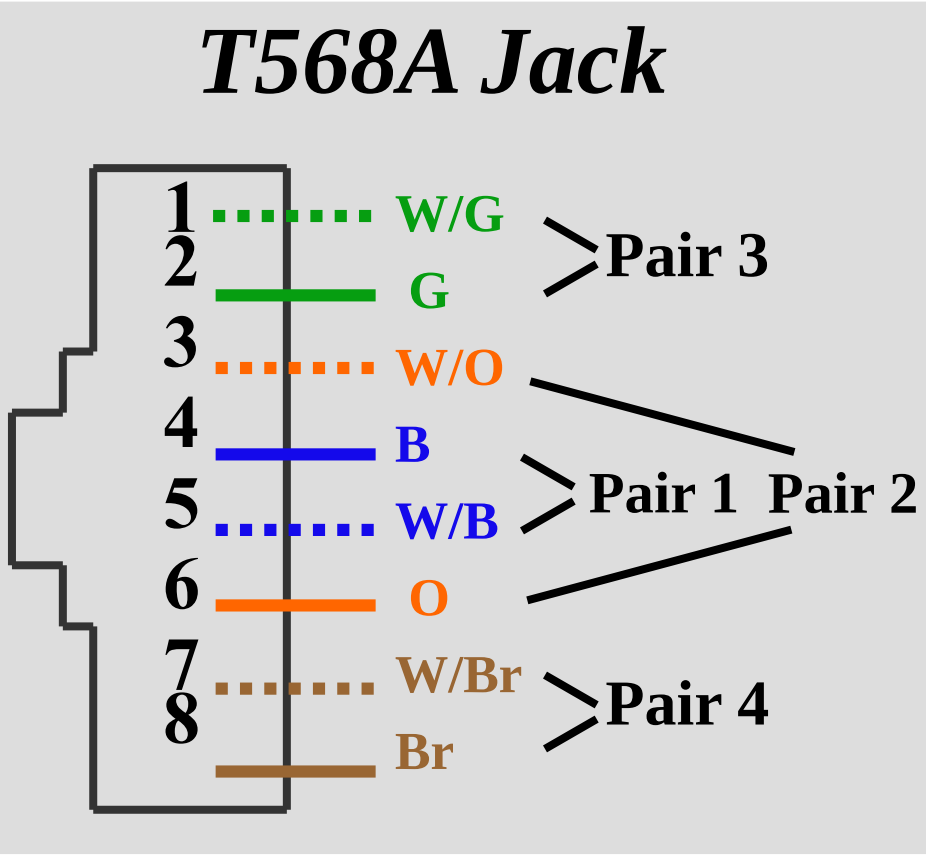


RJ-45 Female

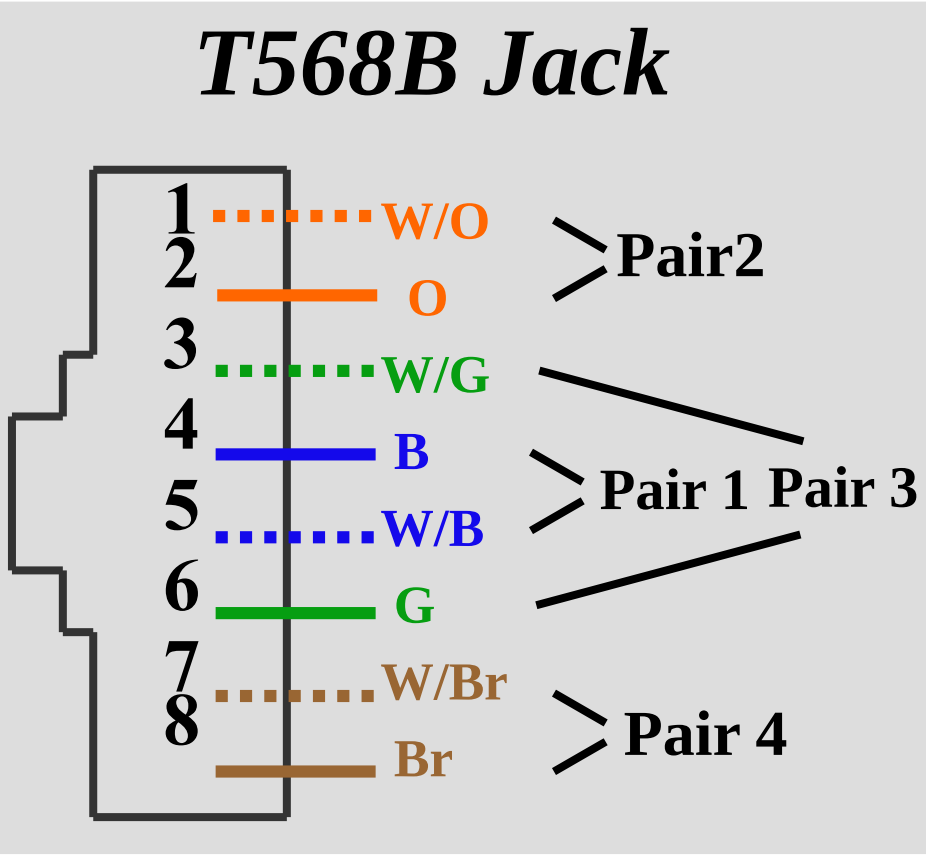


RJ-45 Male

The T568A and T568B wiring patterns

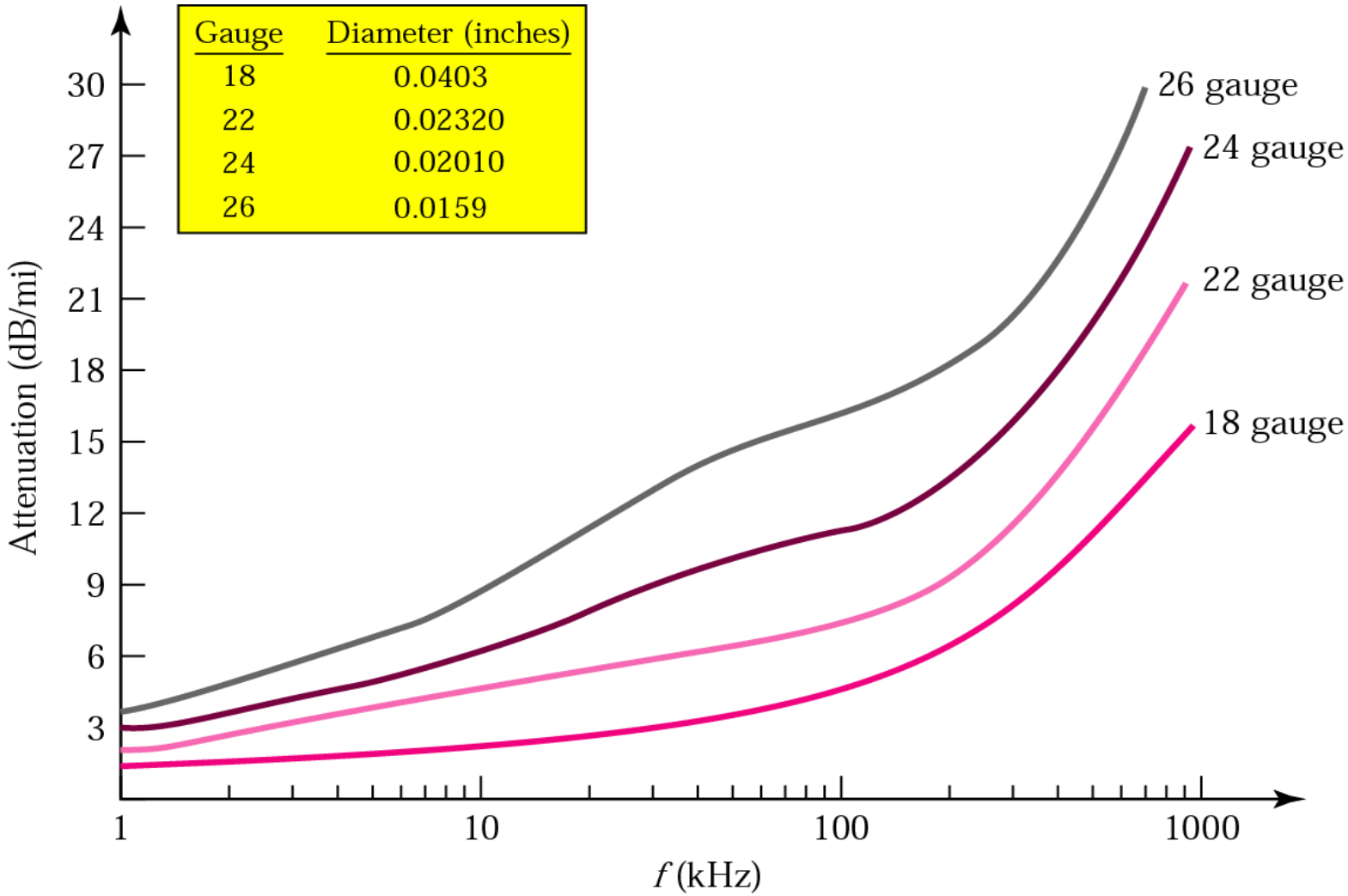


8-pin modular jack
(front view)



8-pin modular jack
(front view)

UTP performance



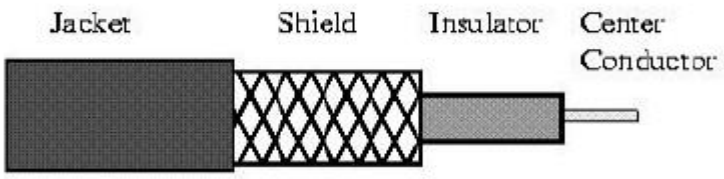
UTP	STP
UTP stands for **Unshielded twisted Pair**	STP stands for **Shielded twisted pair.**
In UTP grounding cable is not necessary	While in STP grounding cable is required.
The data rate in UTP is slow compared to STP.	The data rate in STP is high
The cost of UTP is less.	While STP is costlier than UTP
In UTP much more maintenance is not needed	While in STP much more maintenance is needed
In UTP noise is high compared to STP	While in STP noise is less
short distance such as for home and office networks.	Generally used for connecting organizations over a long distance

Application

- Telephone network(between house and local exchange)
- Private Branch Exchange
- Local Area Network(LAN)

Coaxial Cable

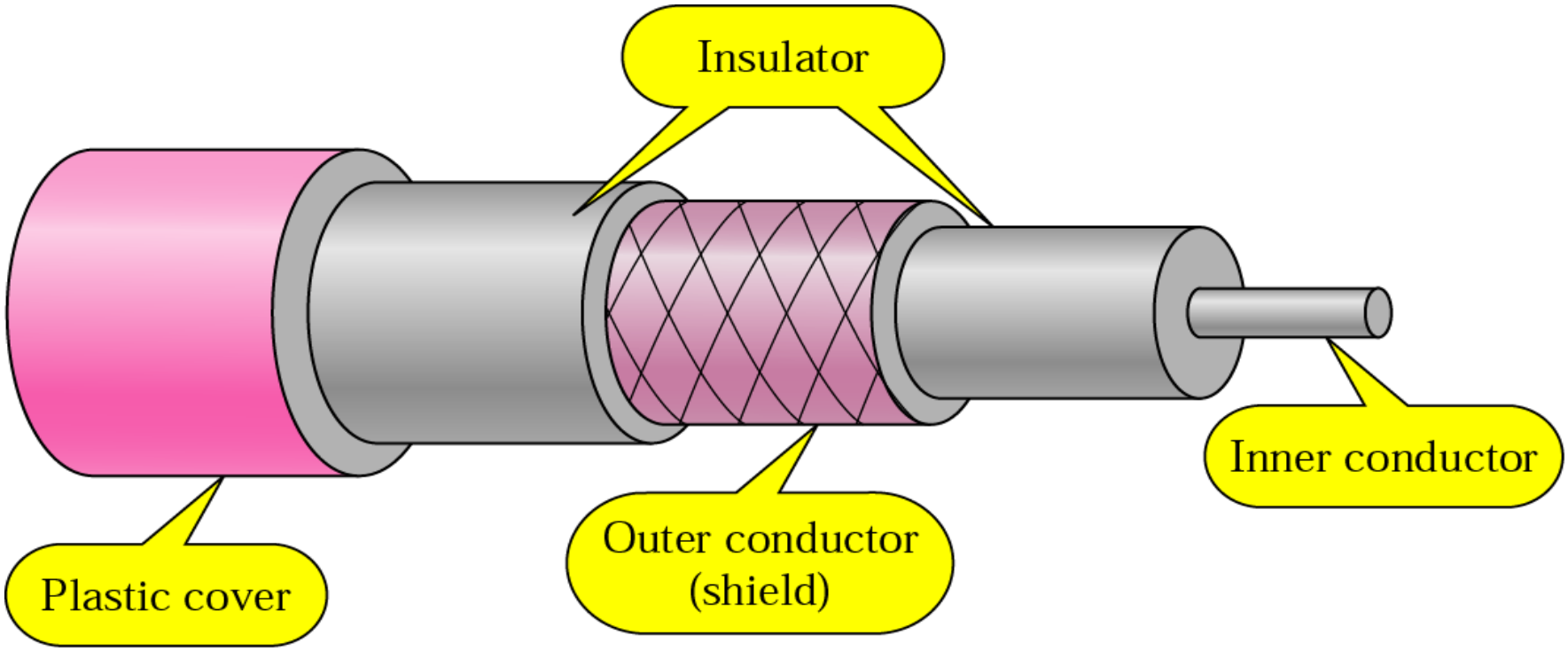
- Has a central core conductor enclosed in an insulating sheath, encased in an outer conductor of metal foil
- Frequency range is between 100 KHz and 500 MHz



Coaxial Cable

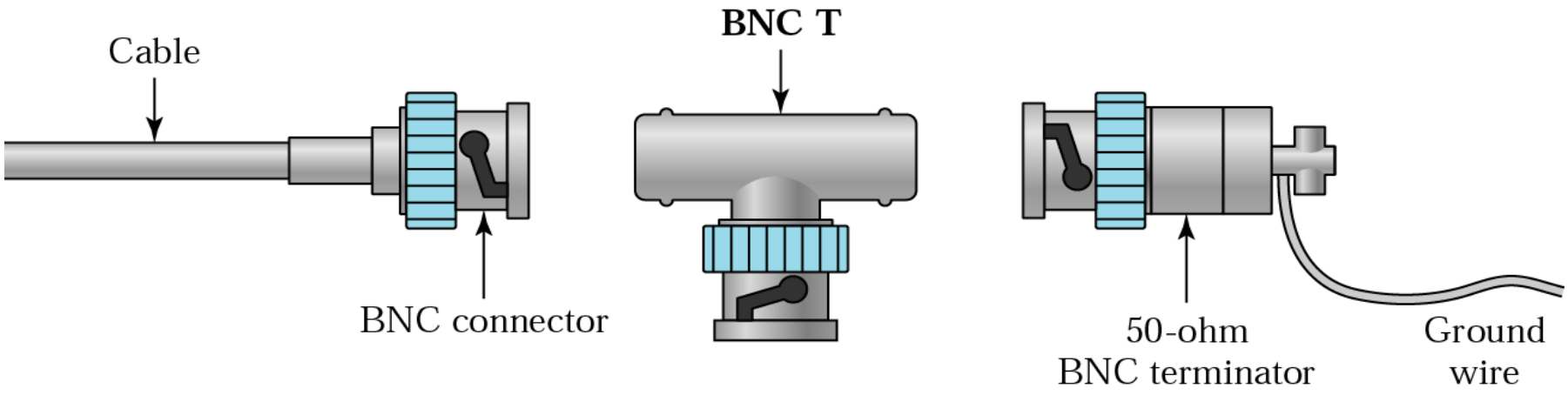
- The outer shield protects the inner conductor from electrical signals.
- Typical impedances for coaxial cables are 75 Ohms for TV cable, 50 Ohms for Ethernet.
- allow higher data rates to be transferred than with twisted pair cable.
- It can be used over longer distance.

Coaxial Cable example



Coaxial Cable Connectors

- Most common is barrel connector (BNC)
- T-connectors are used to branch off to secondary cables



Coaxial Performance

- Higher bandwidth than twisted-pair
- However, attenuation is higher and requires frequent use of repeaters

Coaxial Applications

- Analog telephone networks ,
  - single coaxial network could carry 10,000 voice signals
- Digital telephone networks ,
  - single coaxial network could carry digital data up to 600 Mbps
- Cable TV networks
- Traditional Ethernet LANs

Fiber-Optic Cable

- An fiber-optics is a thin ,flexible medium of guiding an optical ray.
- It consists of three section i.e. Core , cladding & jacket.
- The core is the inner layer and jacket is outer layer . Core is more denser than cladding.

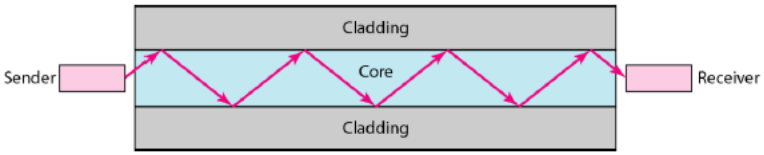
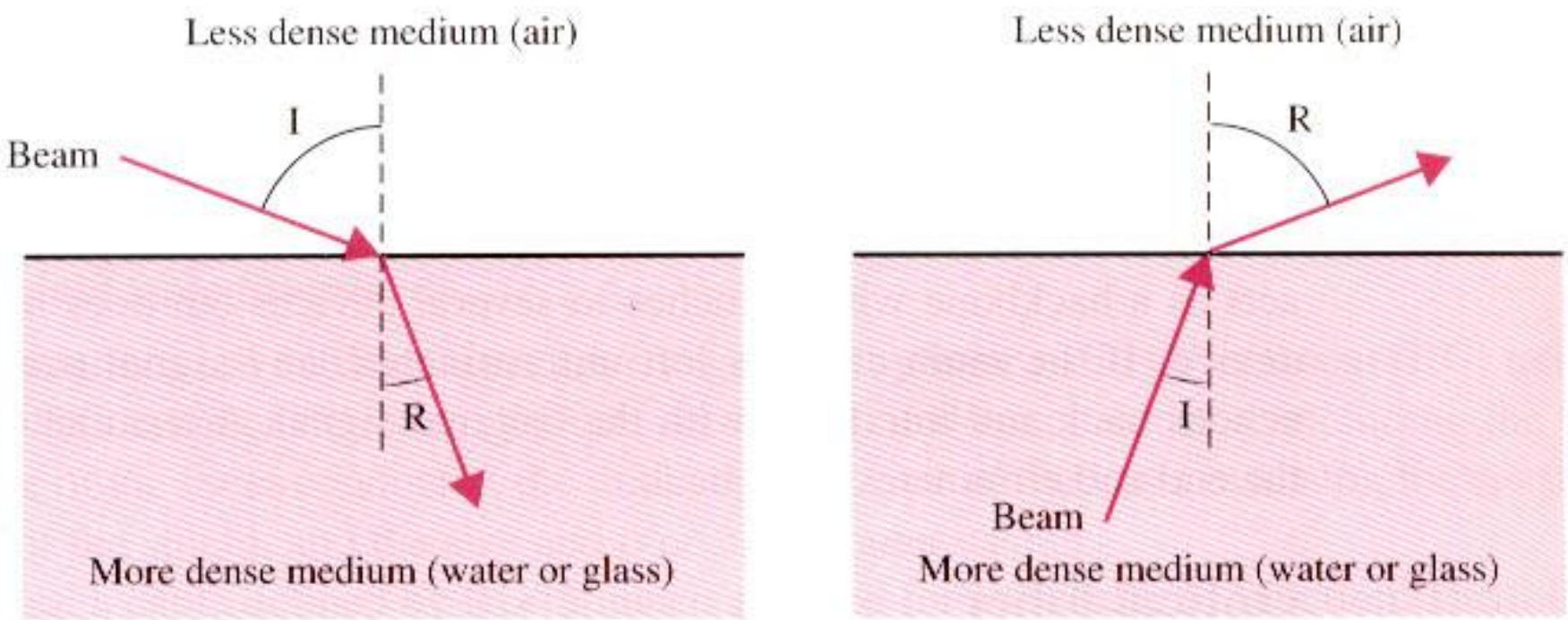


Figure : Optical Fiber

- Made of glass or plastic
  - signals are transmitted as light pulses from an LED or laser
- The interface between the core & cladding acts as a reflector to confined light.
- Typical core diameter is 62.5 microns (1 micron = 10^{-6} m). Typically Cladding has a diameter of 125 microns.
- Speed depends on density of medium it is traveling through;
  - fastest when in a vacuum, 186,000 miles/second

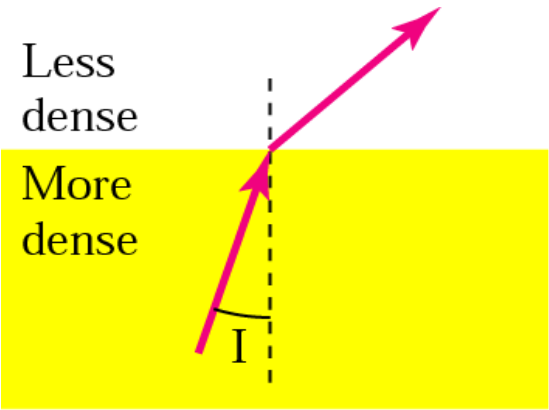
Refraction



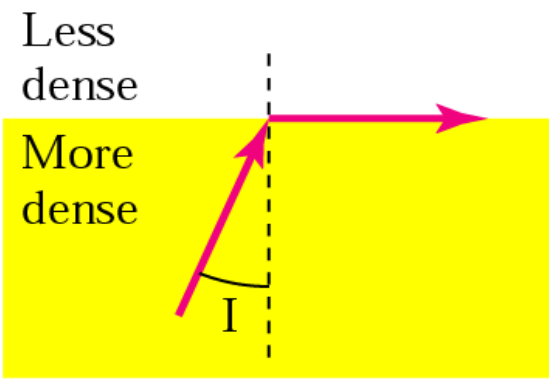
a. From less dense to more dense medium

b. From more dense to less dense medium

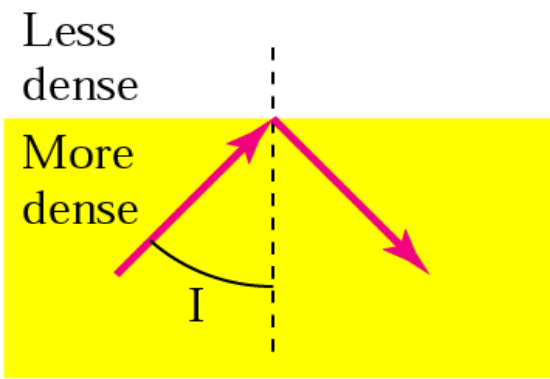
Critical Angle



I < critical angle,
refraction

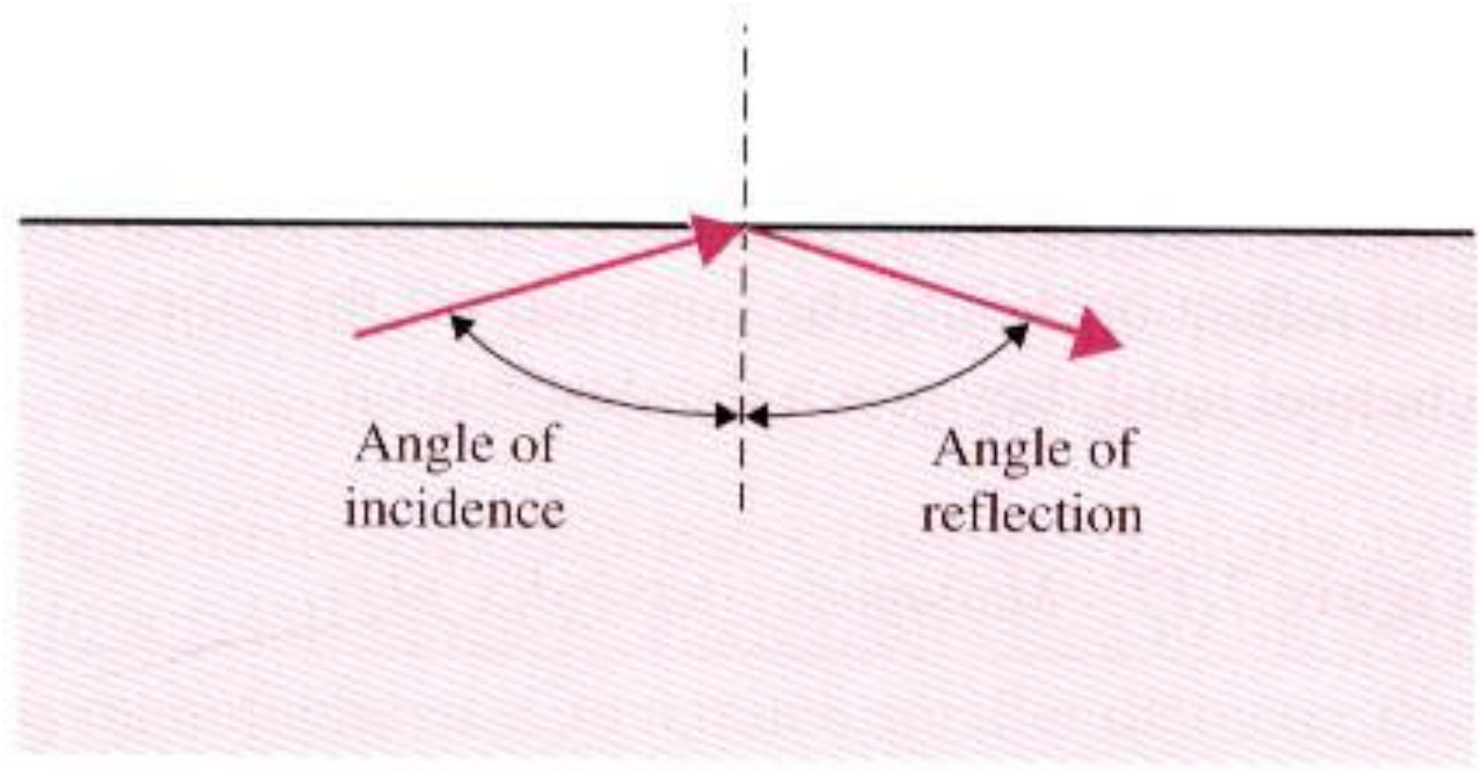


I = critical angle,
refraction

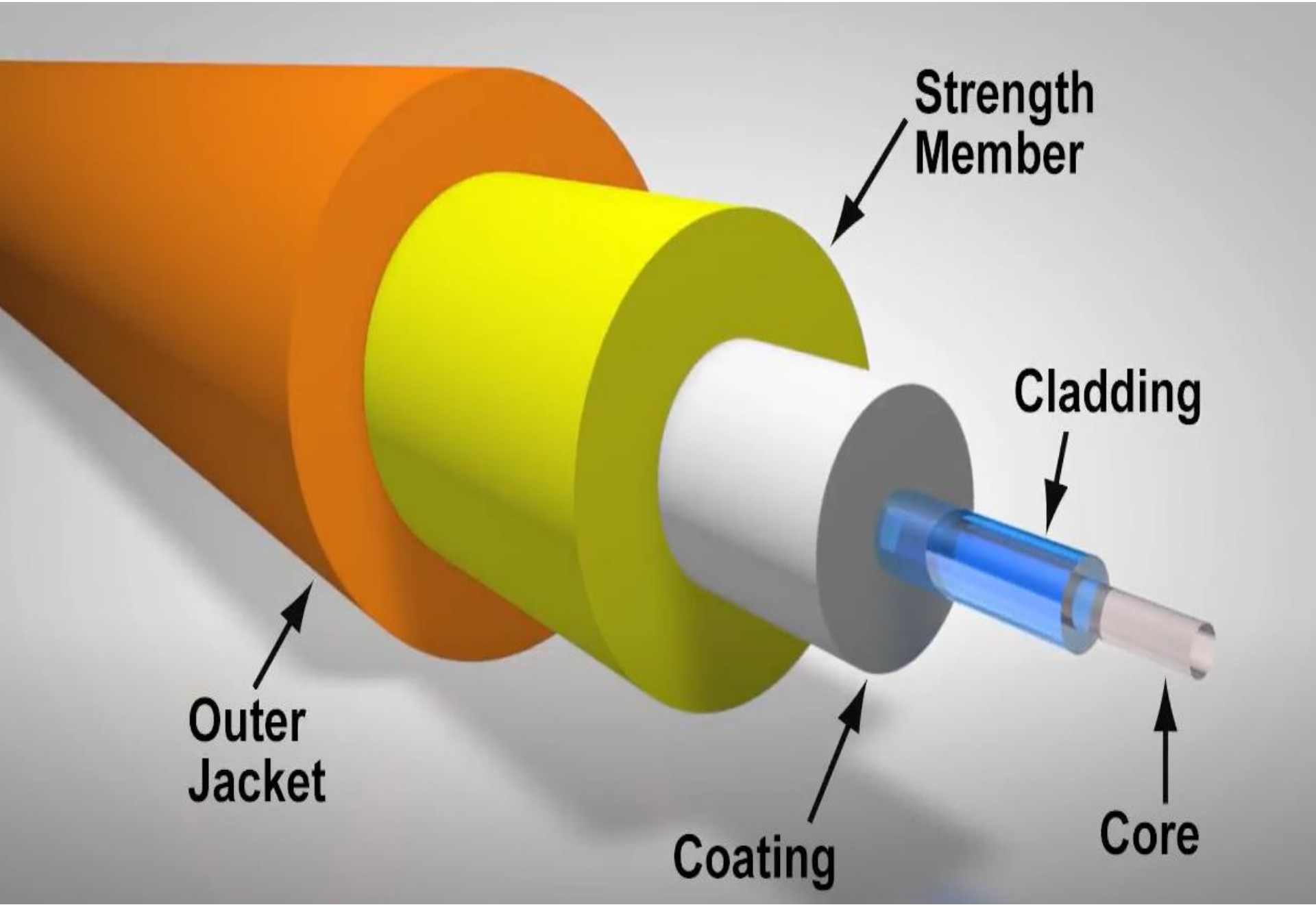


I > critical angle,
reflection

Reflection

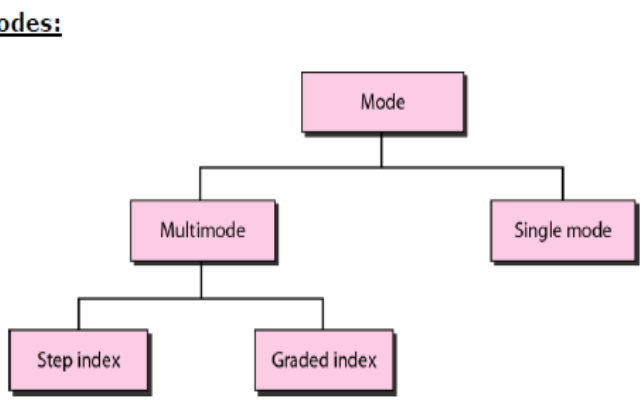


Fiber Construction



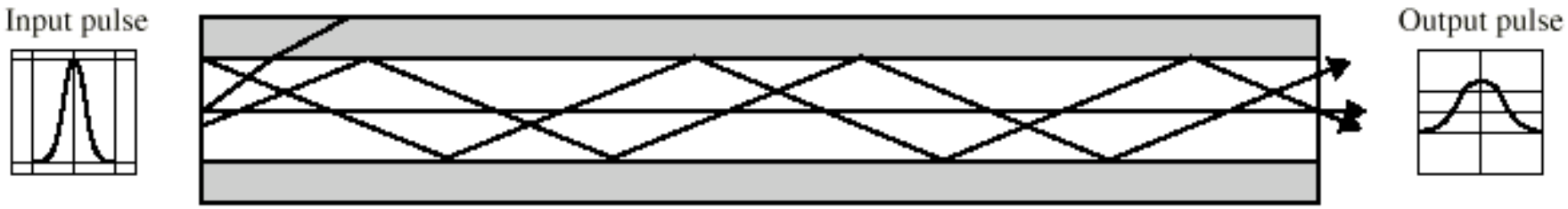
- A **core**, which carries most of the light, surrounded by
- A **cladding**, prevents light from exiting the core which bends the light and confines it to the core
- The **coating**, or buffer, protects the core and cladding and provides strength.
- The next layer is a **strength material** that provides strength to the cable and helps prevent damage due to stress**.**
- The entire package is then encased in a **jacket**. This outer jacket provides one last layer of protection and also adds strength to the fiber.

Propagation Modes

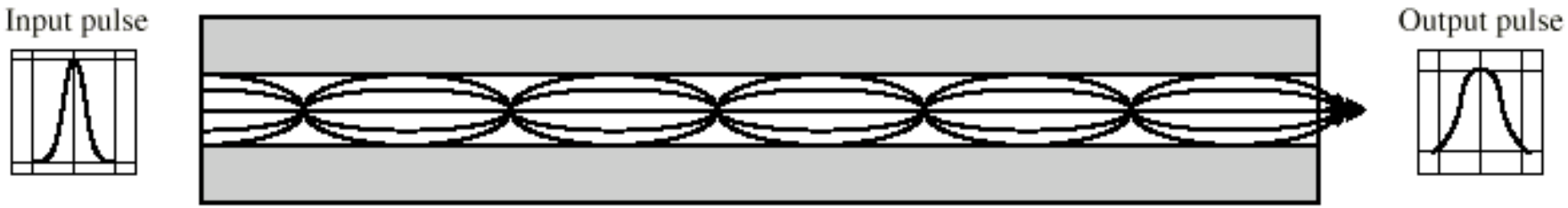


- Method for transmitting optical signals:
  - Multimode
    - Multimode step-index fiber
    - Multimode graded-index fiber
  - Single Mode

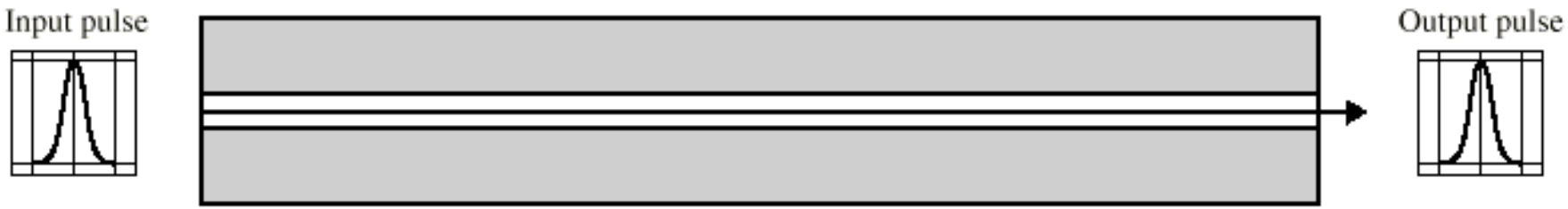
Propagation Modes (cont)



(a) Step-index multimode



(b) Graded-index multimode



(c) Single mode

Multimode

- Multiple beams from light source move through core at different paths
- Multimode step-index fiber
  - *Index* refers to the index of refraction
  - Density remains constant from center to edges
  - Light moves in a straight line until it reaches the cladding
  - Some beams penetrate the cladding and are lost, while others are reflected down the channel to the destination

Multimode (cont)

- As a result, beams reach the destination at different times and the signal may not be the same as that which was transmitted

- Multimode graded-index fiber
  - Second type of fiber called multimode graded index fiber is one with varying densities.
  - Density is highest at center of the core and decreases gradually to its lowest at the edge.

Single Mode

- Single mode uses step index fiber
- Only one beam from a light source is transmitted using a smaller range of angles
- Smaller diameter
- All beams arrive together and can be recombined without signal distortion

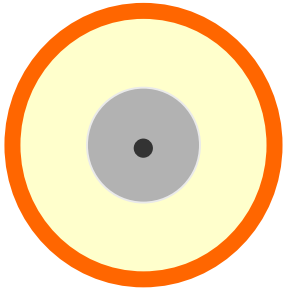
Light Sources

- Light source is light-emitting diode (LED) or a laser
  - LEDs are cheaper but not as precise (unfocused); limited to short-distance use
  - Lasers can have a narrow range, better control over angle
- Receiving device needs a photosensitive cell (photodiode) capable of receiving the signal

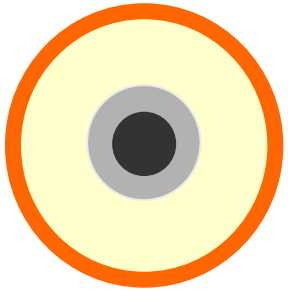
Fiber types

Size	Core	Cladding	Mode
50/125	50	125	Multimode, graded-index
62.5/125	62.5	125	Multimode, graded-index
100/125	100	125	Multimode, graded-index
7/125	7	125	Single-mode

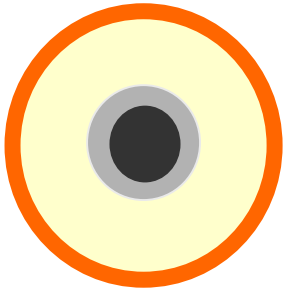
Fiber Sizes



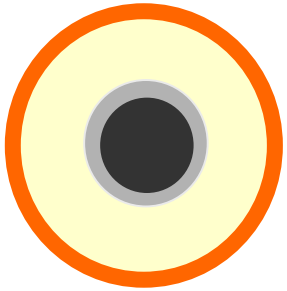
Single mode
7/125



Multimode
50/125

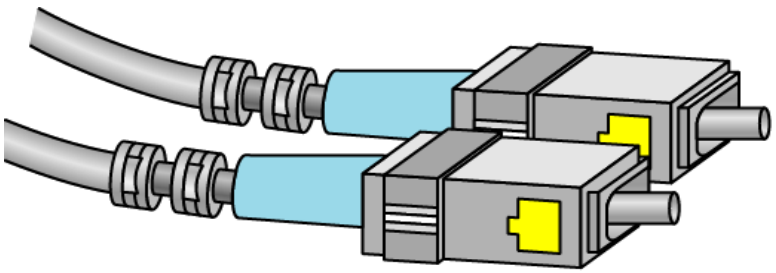


Multimode
62.5/125

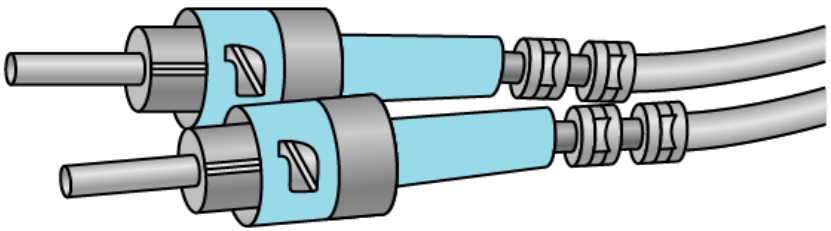


Multimode
100/125

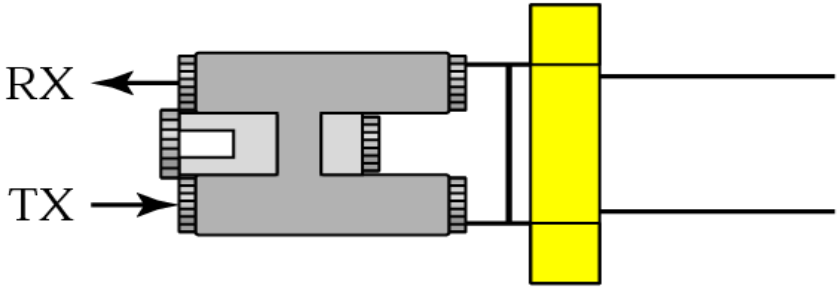
Fiber-Optic cable connectors



SC connector

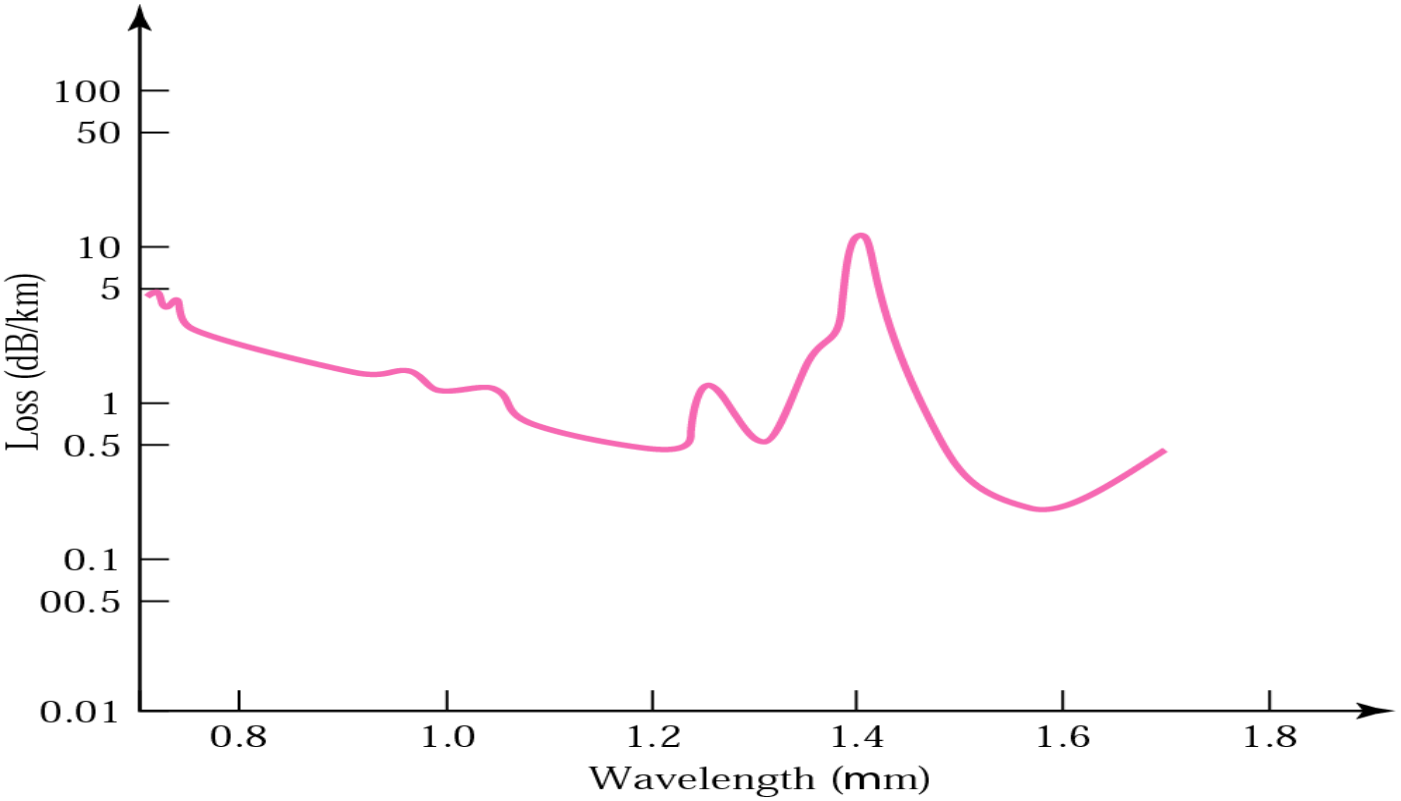


ST connector



MT-RJ connector

Optical fiber performance



Optical fibers transmit data using specific wavelengths of light, typically in the infrared spectrum, with **850 nm, 1300 nm, and 1550 nm being the most common.**

- **LED (Light Emitting Diode) light source**
- 3.5 dB/Km Attenuation (loses 3.5 dB of signal per kilometer)
- 850 nM - wavelength of light source

- **Laser Light Source**
- 1 dB/Km Attenuation (loses 1 dB of signal per kilometer)
- 1170 nM - wavelength of light source

Applications of Fiber Optics

- Telecommunications and Networking: **Telephone and Cable TV, Data Centers**
- Backbone networks due to wide bandwidth and cost effectiveness
- Medical Applications: **Endoscopy, Medical Imaging, Surgical Lasers**
- Industrial Applications
- Decorative and Lighting
- Up to 1600 Gbps with WDM

Advantages of Fiber Optics

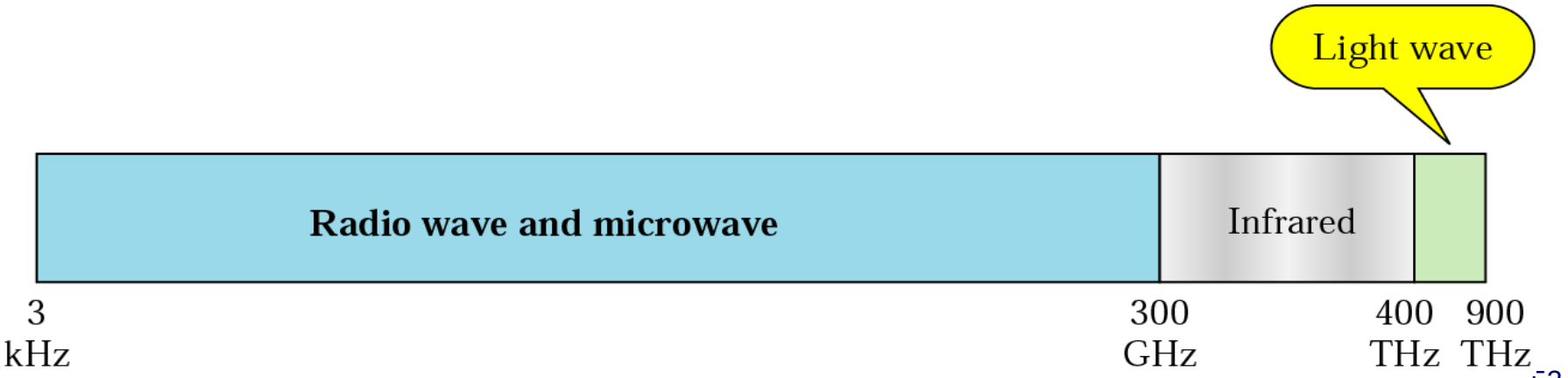
- Lightweight
- Higher bandwidth than twisted-pair and coaxial cable.
- Security: cannot tap into cable
- Less signal attenuation
- Immunity to EMI
- More resistant to corrosive materials
- Longer distances than copper wire
- Faster transmission rate

Disadvantages of Fiber Optics

- Installation/maintenance
  - Relatively new technology
  - need expertise
- Unidirectional, bidirectional communication two fibers are needed
- Cost
  - Cable and interface are relatively more expensive

Unguided Media: Wireless

- Wireless communication: transporting electromagnetic waves without a physical conductor.
- They are not guided or bound to a channel to follow.
- They are classified by the type of wave propagation .
- Instead, they rely on **electromagnetic waves** transmitted through air, vacuum, or water. These are commonly used in wireless communication systems.



Characteristics of Unguided Communication Bands

- **Wireless Transmission :** Data is transmitted through electromagnetic waves, such as radio waves, microwaves, and infrared waves, without the need for physical cables.
- **Long-Distance Coverage :** Some unguided media, like radio waves, can cover vast distances, allowing for communication across long terrains.
- **High Bandwidth :** Unguided media, particularly microwaves, can offer a wide range of bandwidth, enabling high data transfer speeds.

  -Radio Waves: 3 kHz to 1 GHz

  -Microwaves: 1 GHz to 300 GHz

  -Infrared: 300 GHz to 400 THz
- **Directionality:**

-Radio waves: Omnidirectional (spread in all directions).

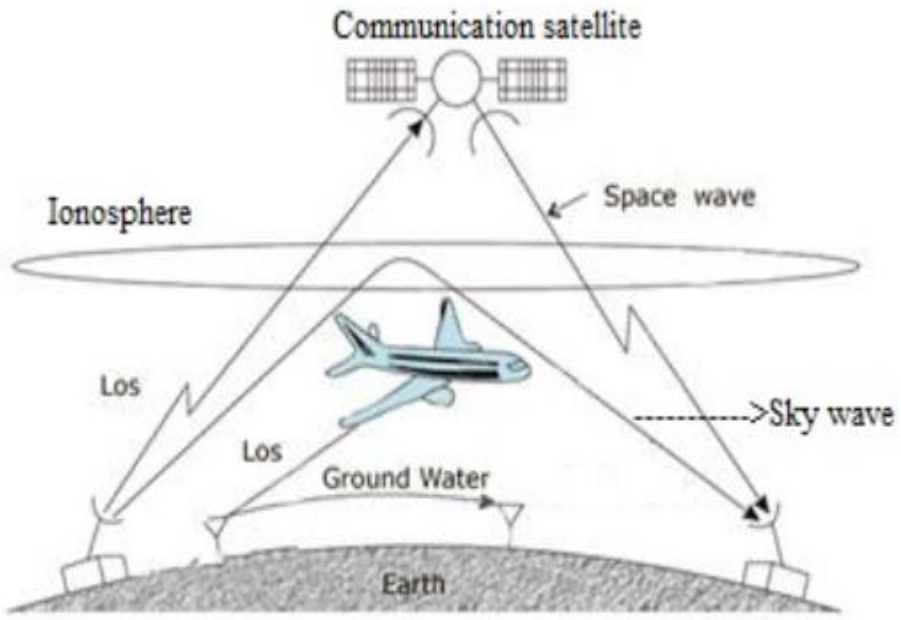
-Microwaves and Infrared: More directional, requiring line-of-sight

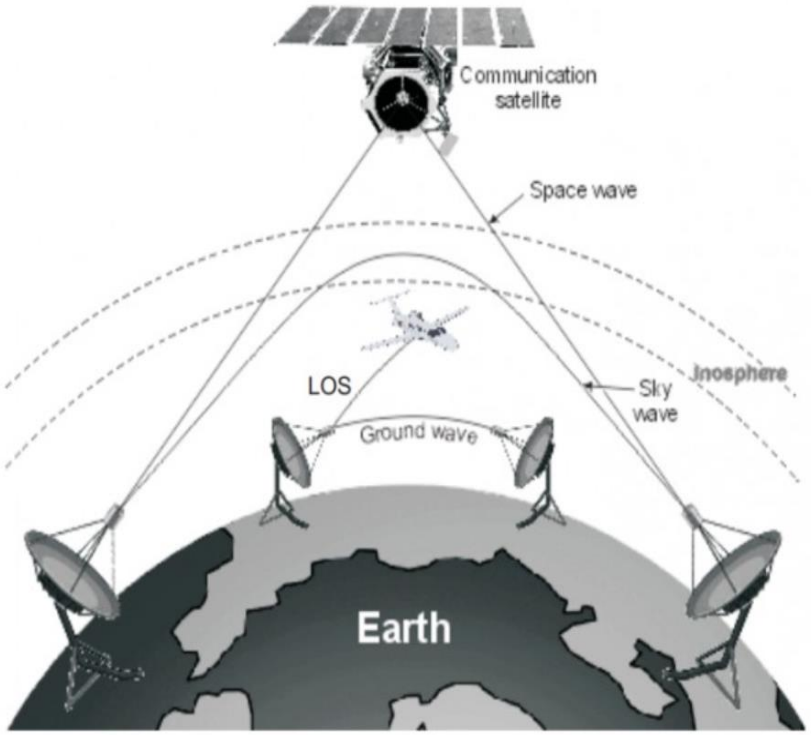
- **Security Considerations :** Unguided media can be more susceptible to interception and interference compared to guided media due to their nature of being broadcast through the air.
- **Diverse Applications:** They are used in a wide array of applications, including mobile communication (cellular phones), Bluetooth, satellite communication, wireless local area networks (Wi-Fi), and more

Wireless Propagation Methods

- Ground – radio waves travel through lowest portion of atmosphere, hugging the Earth
- Sky – higher-frequency radio waves radiate upward into ionosphere and then reflect back to Earth
- Line-of-sight – high-frequency signals transmitted in straight lines directly from antenna to antenna

Propagation Methods



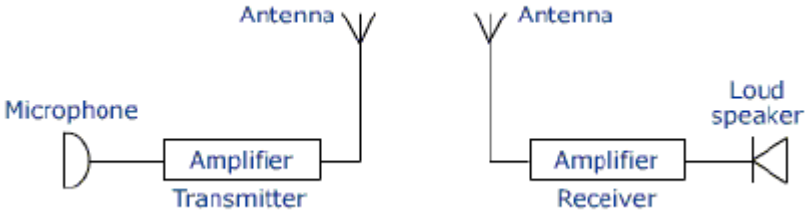


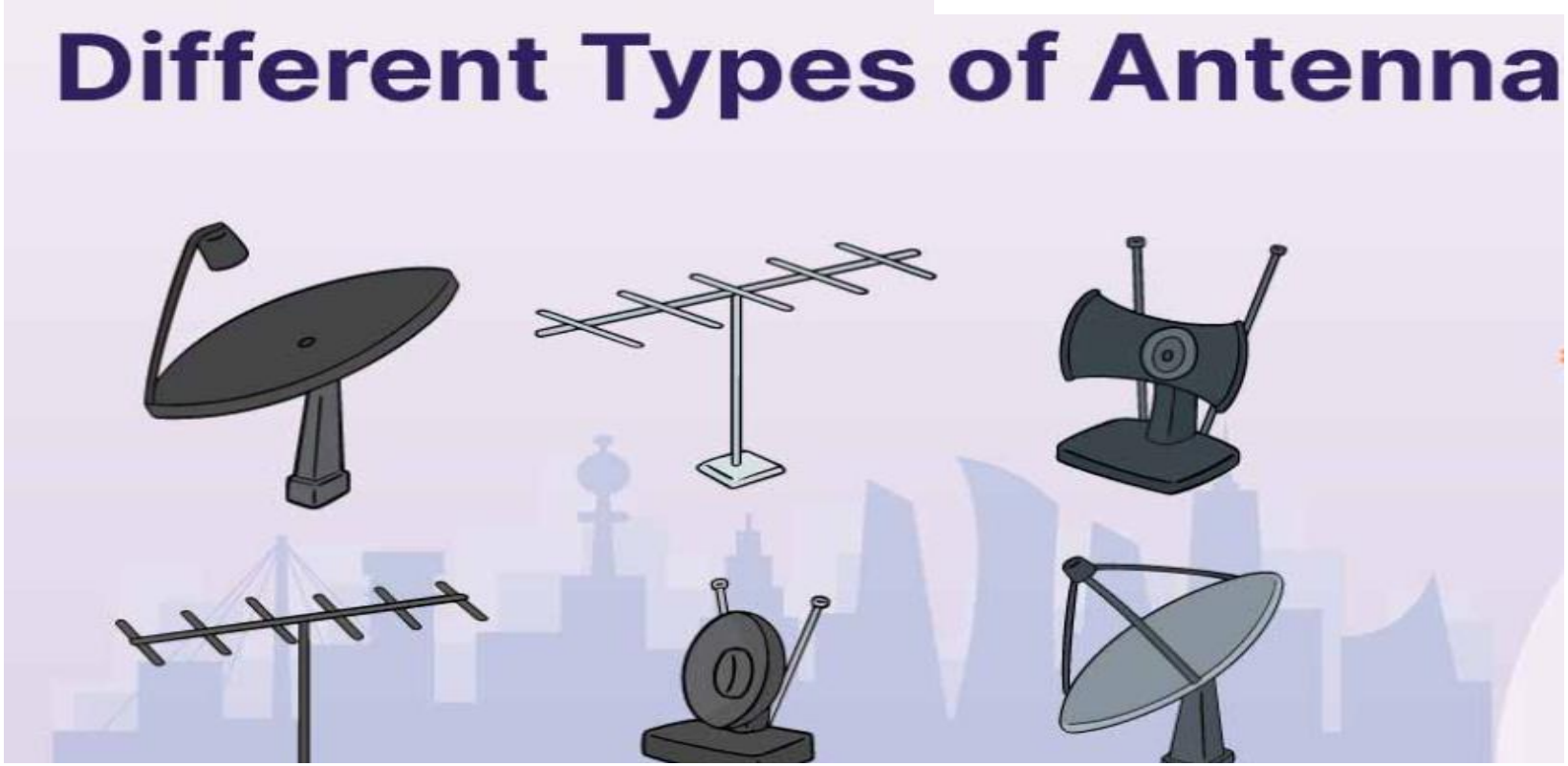
Bands

Band	Range	Propagation	Application
VLF	3–30 KHz	Ground	Long-range radio navigation
LF	30–300 KHz	Ground	Radio beacons and navigational locators
MF	300 KHz–3 MHz	Sky	AM radio
HF	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF	3–30 GHz	Line-of-sight	Satellite communication
EHF	30–300 GHz	Line-of-sight	Long-range radio navigation

Antenna

- An antenna is defined as an electrical conductor or system of conductors used either for radiating electromagnetic energy or for receiving electromagnetic energy.





Wireless Transmission Waves

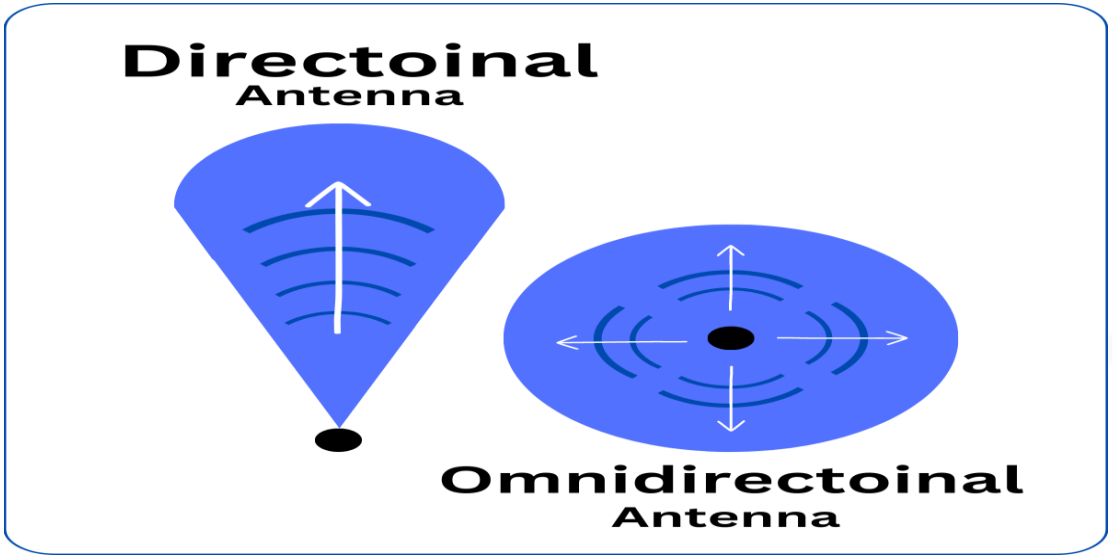
- Radio Waves
- Microwave
- Infrared

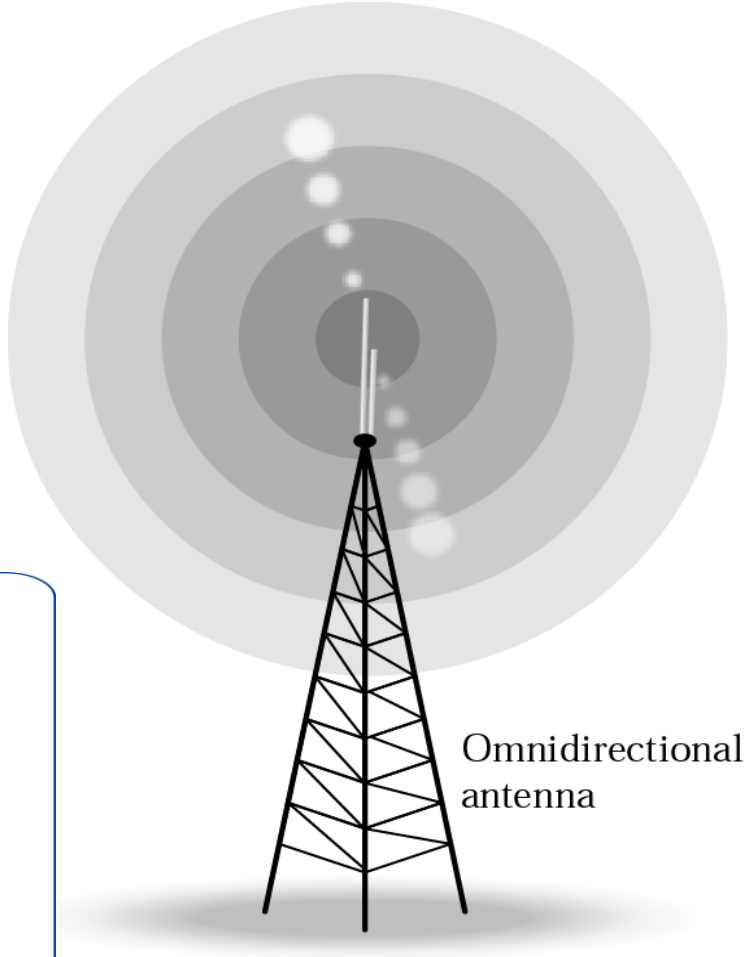
Radio Waves

- Frequency ranges: 3 KHz to 1 GHz
- Omnidirectional
- Susceptible to interference by other antennas using same frequency or band
- Ideal for long-distance broadcasting
- May penetrate walls
- Apps: AM and FM radio, TV, maritime radio, cordless phones, paging

Omnidirectional antennas

As the name suggests, **omnidirectional antennas** receive signals equally well from all directions. This means they have a 360-degree reception pattern and are generally omnidirectional in the vertical and horizontal planes.

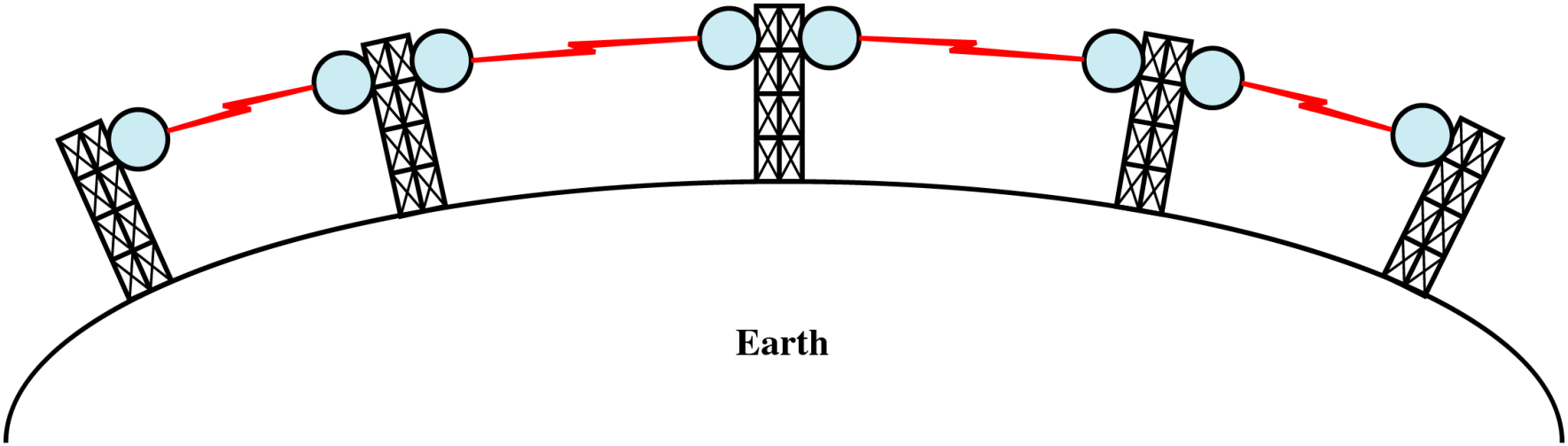




Microwaves

- Frequencies between 1 and 300 GHz
- Unidirectional
- Narrow focus requires sending and receiving antennas to be aligned
- Issues:
  - Line-of-sight (curvature of the Earth; obstacles)
  - Cannot penetrate walls

Terrestrial Microwave



Microwave Applications

- Unicasting – one-to-one communication between sender and receiver
  - Cellular phones
  - Satellite networks
  - Wireless LANs

- Microwaves operate at high operating frequencies of 3 to 10 GHz. This allows them to carry large quantities of data due to their large bandwidth.

Advantages:

a. They require no right of way acquisition between towers.

b. They can carry high quantities of information due to their high operating frequencies.

c. High frequency/short wavelength signals require small antennae.

Disadvantages:

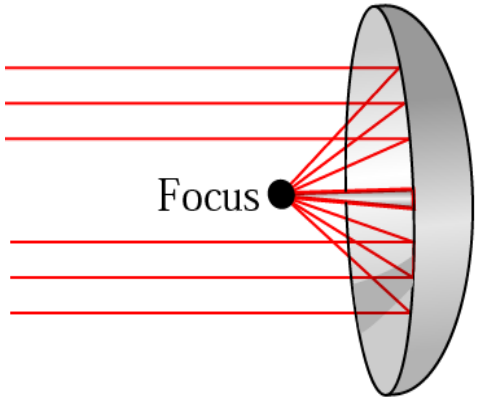
a. Attenuation by solid objects: birds, rain, snow and fog.

b. Diffracted (split) around solid objects.

c. Refracted by atmosphere, thus causing beam to be projected away from receiver.

Parabolic Dish Antenna

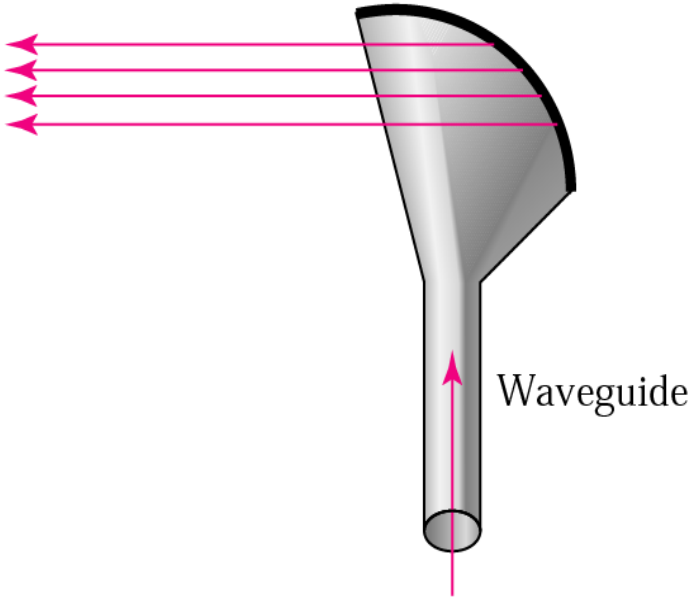
- Incoming signals - Signal bounces off of dish and is directed to focus
- Outgoing signals – transmission is broadcast through horn aimed at dish and are deflected outward



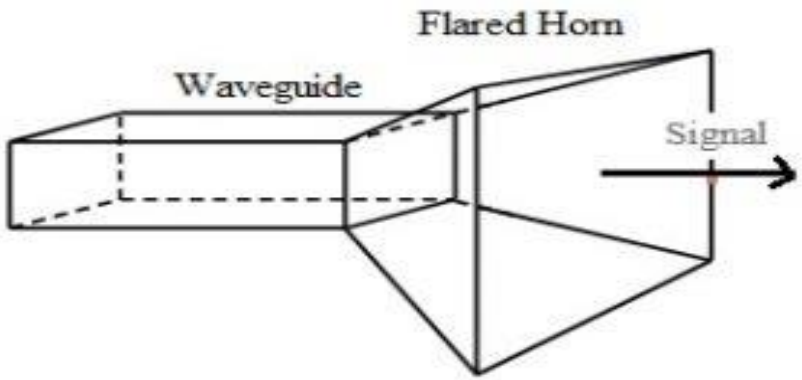
a. Dish antenna

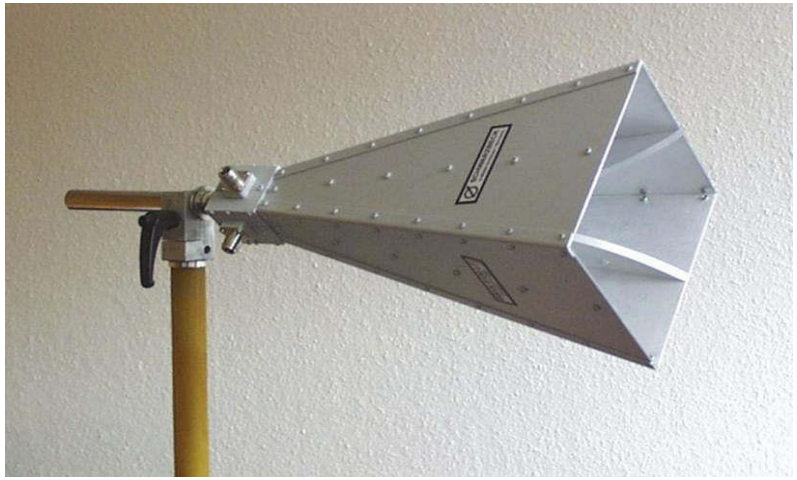
Horn Antenna

- Outgoing transmissions broadcast through a stem and deflected outward
- Received transmissions collected by a scooped part of the horn and deflected downward into the stem
- Horn antennas are widely used in satellite communication ground stations, radar systems, and measurement systems due to their high gain and directional radiation patterns.



b. Horn antenna





Satellite Microwave

- Similar to terrestrial microwave except the signal travels from a ground station on earth to a satellite and back to another ground station.
- Satellite receives on one frequency, amplifies or repeats signal and transmits on another frequency
- Satellite is relay station
- Television
- Long distance telephone
- Private business networks

- A satellite is simply anybody that moves around another body (usually much larger) one in a mathematically predictable path called an orbit. eg Intelsat, GSAT 30, GSAT 20 etc.
- Satellite communication is the method of **transporting information from one place to another using a communication satellite** in orbit around the Earth.
- A communication satellite is an artificial satellite that transmits the signal via a **transponder by creating a channel between the transmitter and the receiver** located at different locations on the Earth.

Satellite Orbit:

An orbit is the curved path that an object in space (such as a star, planet, moon, asteroid or spacecraft) takes around another object due to gravity.

Types of Orbit

1.LEO(Low earth Orbit)

LEO satellites are much closer to the earth than GEO satellites, ranging from 500 KM to 1500 km above earth surface. visible for 15 to 20 minutes

2.MEO(Medium Earth Orbit)

A MEO satellite is in orbit somewhere between 8000 km to 18000 km above from the earth surface. Visible for much longer periods of time than LEO satellites, usually between 2 to 8 Hours.

3.GEO(Geostationary Orbit)

If the satellite rotating in this orbit has the angular velocity equal to the earth's angular velocity, then the satellite is said to be moving along with earth. It appears to be stationary to the point of the earth. Orbit is somewhere 35,786 kilometers above earth surface

4.Polar Orbit and Sun synchronous Orbit

A Polar Orbit (PO) is an orbit in which a satellite passes above or nearly above Earth's North and South poles on each revolution.

The communication satellites are similar to the **space mirrors that help us in bouncing the signals** such as radio, internet data, and television from one side of the earth to another. There are three stages that are involved which explain the working of satellite communications. These are:

- Uplink
- Transponders
- Downlink

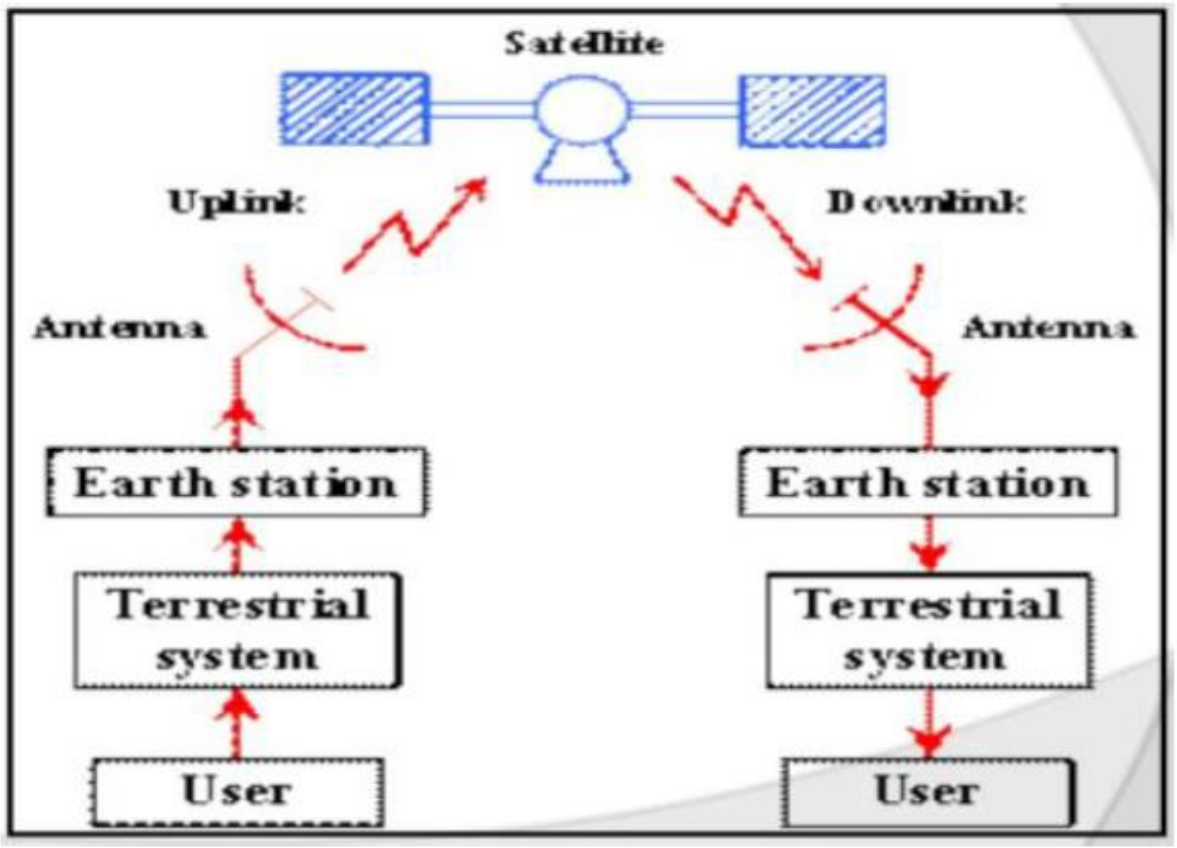
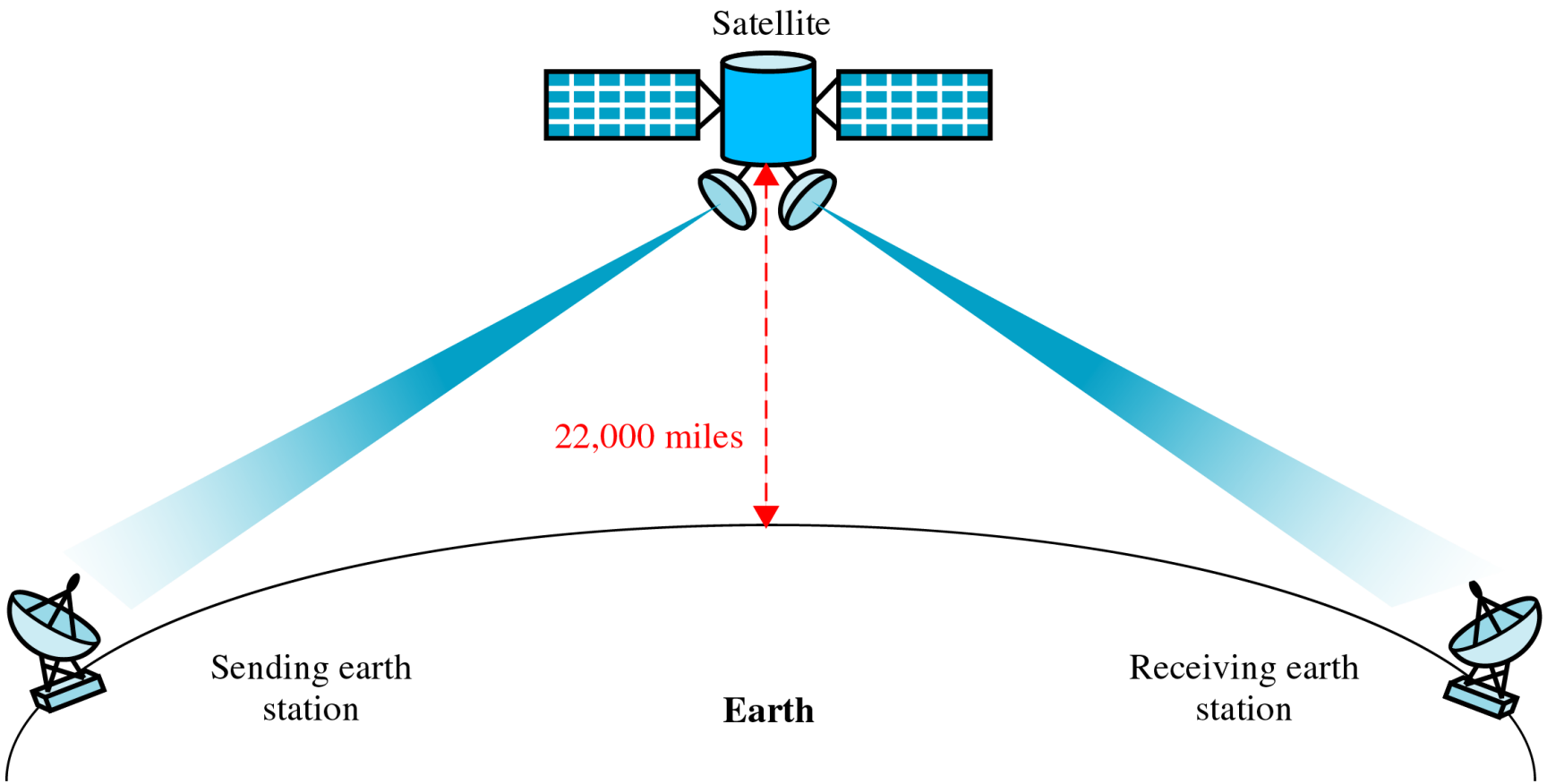


Figure: Fundamental of Satellite Communication

Satellite as a repeater



Satellite frequency band

Band	Downlink (GHz)	Uplink (GHz)	Bandwidth, (MHz)
C	3.7 to 4.2 GHz	5.925 to 6.425 GHz	500
Ku	11.7 to 12.2 GHz	14 to 14.5 GHz	500
Ka	17.7 to 21 GHz	27.5 to 31 GHz	3500

Advantages of satellite Communication

- **Global Coverage**
- **Flexibility and Mobility**: allowing for quick deployment and redeployment of communication system
  used for mobile communication on ships, aircraft, and other moving vehicles,
- Satellite communication can be a **cost-effective** solution for long-distance communication and broadcasting
- **Reliability and Speed**

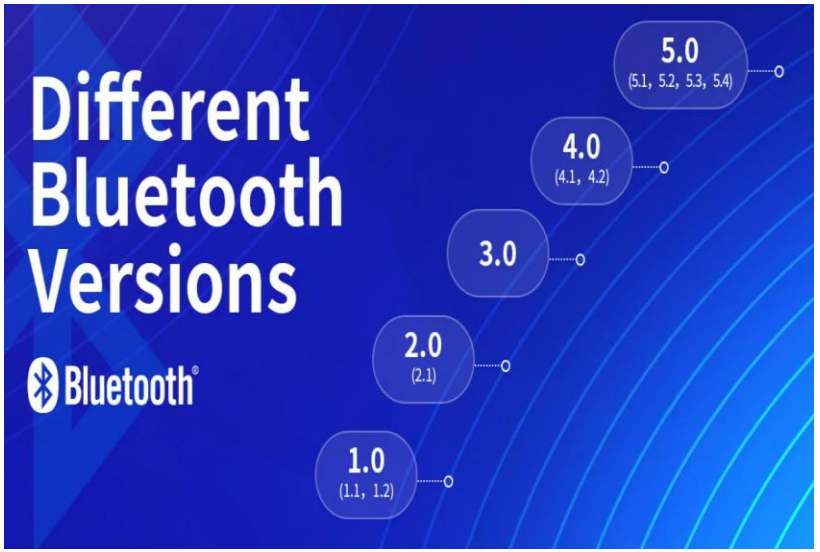
Disadvantages

- The initial investment in satellite is **costlier**
- Satellites have a finite capacity for data transmission, which can lead to **congestion and slower speeds**
- **Weather Dependency**
- **Latency** :*The time it takes for signals to travel to and from a satellite can introduce delays, making it unsuitable for real-time applications*
- **Security Concerns**

Bluetooth(for knowledge)

- Bluetooth is a Radio Frequency specification for short-range, point-to-multipoint voice and data transfer.
- Bluetooth can transmit through solid, non-metal objects.
- Its typical link range is from 10 cm to 10 m, but can be extended to 100 m by increasing the power.
- **Frequency Band**: **2.4 GHz ISM band** (Industrial, Scientific, and Medical band). Divided into **79 channels**, each **1 MHz** wide (in classic Bluetooth)
- Frequency Hopping spread spectrum

Frequency hopping means that a Bluetooth device ***rapidly changes the frequency*** *it uses to transmit data. Instead of using one frequency channel continuously, it switches among many predefined channels in a* ***pseudo-random sequence***



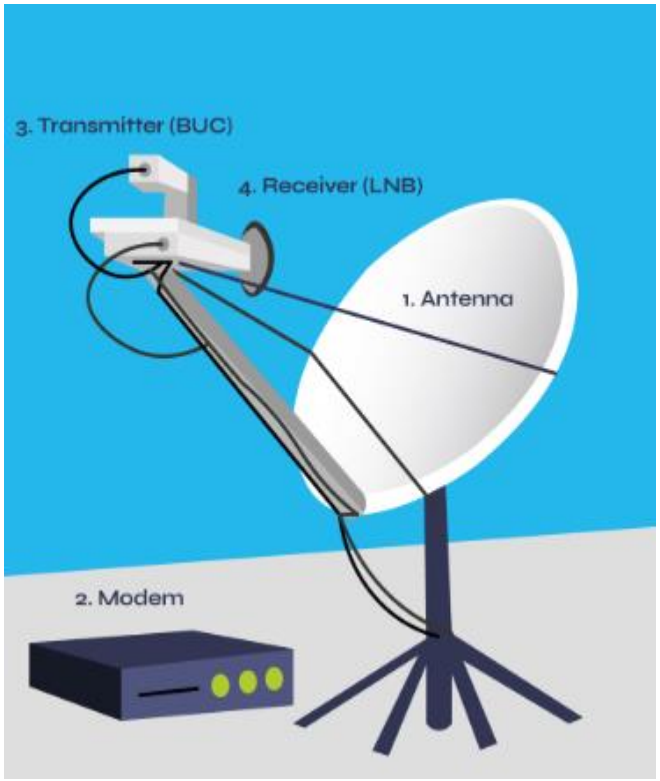
Bluetooth

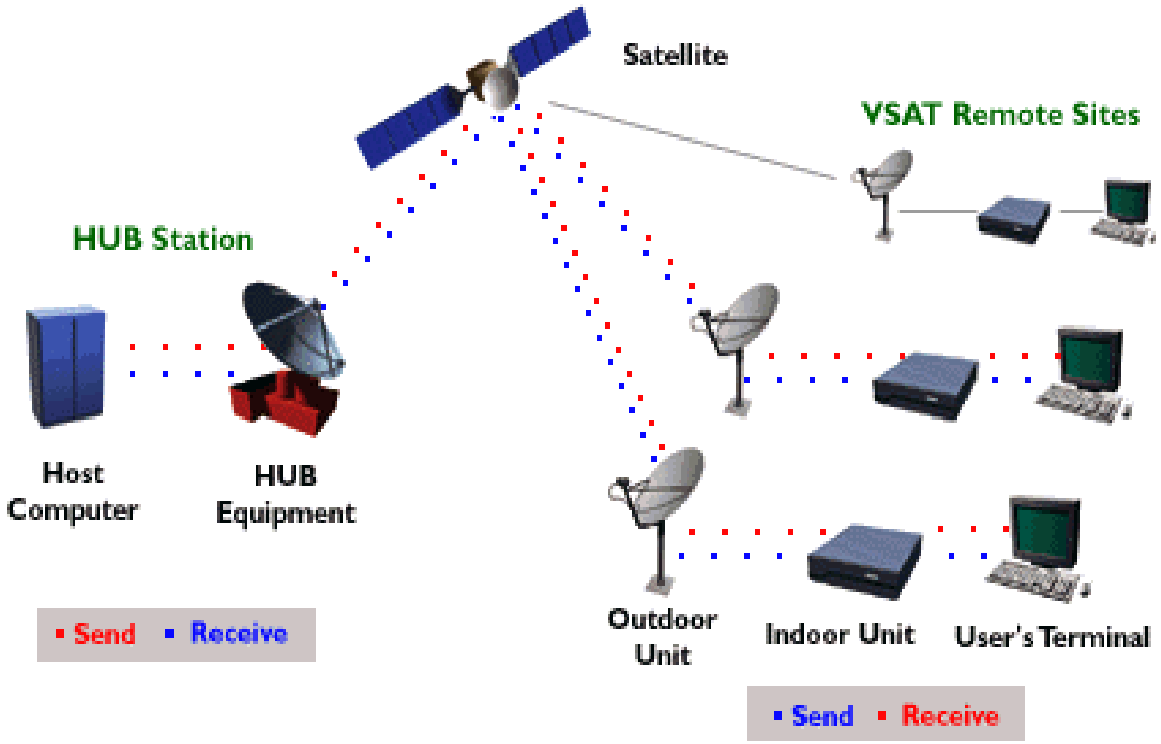
- Bluetooth will enable users to connect to a wide range of computing and telecommunication devices without the need of connecting cables.
- Typical uses include phones and pagers, modems, LAN access devices, headsets, notebooks, desktop computers.

Very Small Aperture Terminal (VSAT)

- A very small aperture terminal (VSAT) is a small-sized earth station used in the transmit/receive of data, voice and video signals over a satellite communication network.
- The satellite provider can divide the total capacity into a number of channels and lease these channel to individual business users.A user equipped with the antenna at a number of sites can use a satellite channel for a private network.
- A VSAT consists of two parts: a transceiver placed outdoors in direct line of sight to the satellite, and a device that is placed indoors to interface the transceiver with the end user's communications device, such as a PC.
- VSAT is a technology that is used for effective communication in remote areas using Satellites and components located on ground.
- VSATs (Very Small Aperture Terminals) is a two way, lost cost, ground micro station for transmitting data to and from communication satellites

- A VSAT has a dish antenna with diameters between 75 cm to 1 m, which is very small in comparison with 10 m diameter of a standard GEO antenna .
- Data rates in VSATs ranges from 4 Kbps to 16 Mbps.





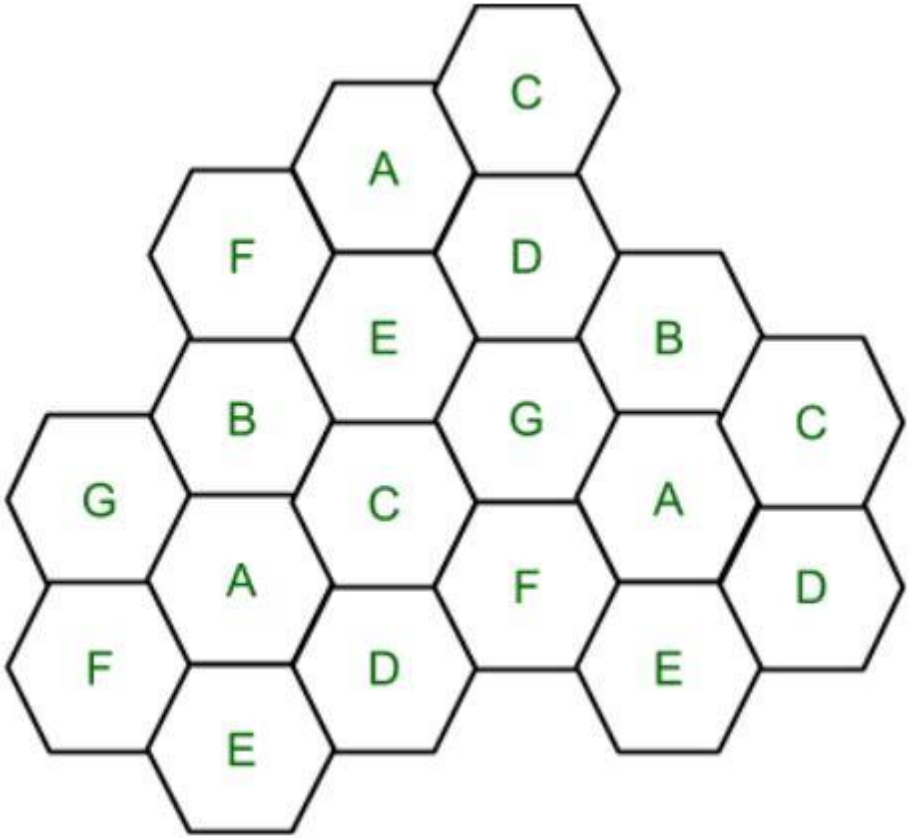
VSAT –Application

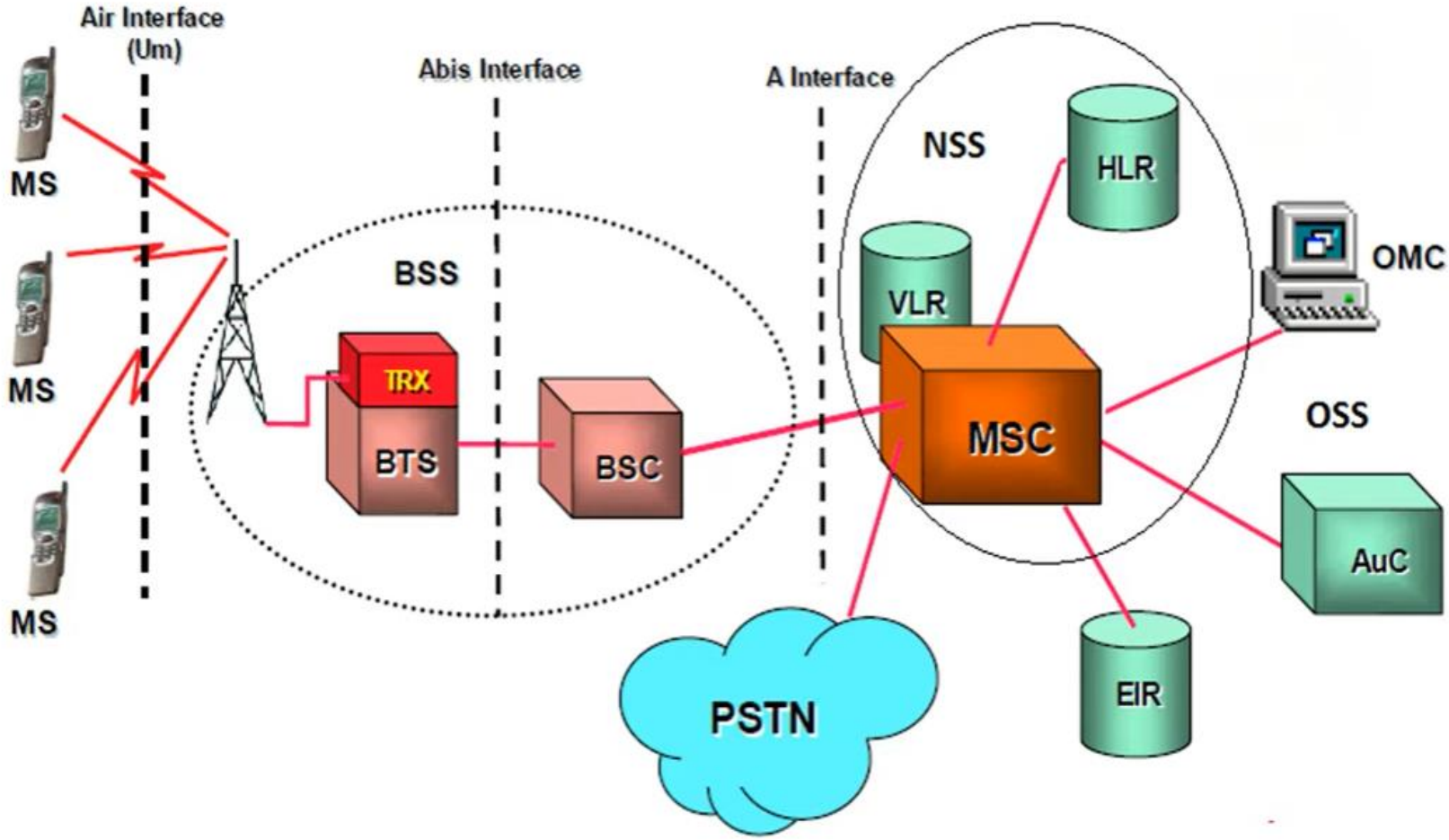
- ✓ **Banking and Financial Services:**
- ▪ **ATM connectivity** in remote or rural areas
- ▪ Secure **transaction processing** and **data transfer** between branches
- ✓ **Enterprise Networks:**
- ▪ Connects **corporate offices**, warehouses, and factories across geographies
- ✓ **Government and Military:**
- ▪ **Disaster recovery** and **emergency communications**
- ▪ Remote border and field communications
- ✓ **Education:**
- ▪ **E-learning centers** in rural areas
- ✓ **Telemedicine:**
- ▪ Supports **video consultations**, remote diagnostics, and transmission of medical data in underserved areas

Cellular Telephony

- A cellular telephone system provides a wireless connection to the terrestrial telephone network (PSTN: Public switch Telephone Network) for any user location within the radio range of the system.
- Cellular systems accommodate a large number of users over a large geographic area, within a limited frequency spectrum.
- Cellular radio system provides high quality service that is often comparable to that of the landline telephone systems.
- High capacity is achieved by limiting the coverage of each base station transmitter to a small geographic area called a cell so that the same radio channels may be reused by another base station located some distance away.
- A sophisticated switching technique called a handoff enables a call to proceed uninterrupted when the user moves from one cell to another.

- Geographic service is divided into smaller "Cells".
- Neighbouring cells do not use name set of frequencies to prevent interference.





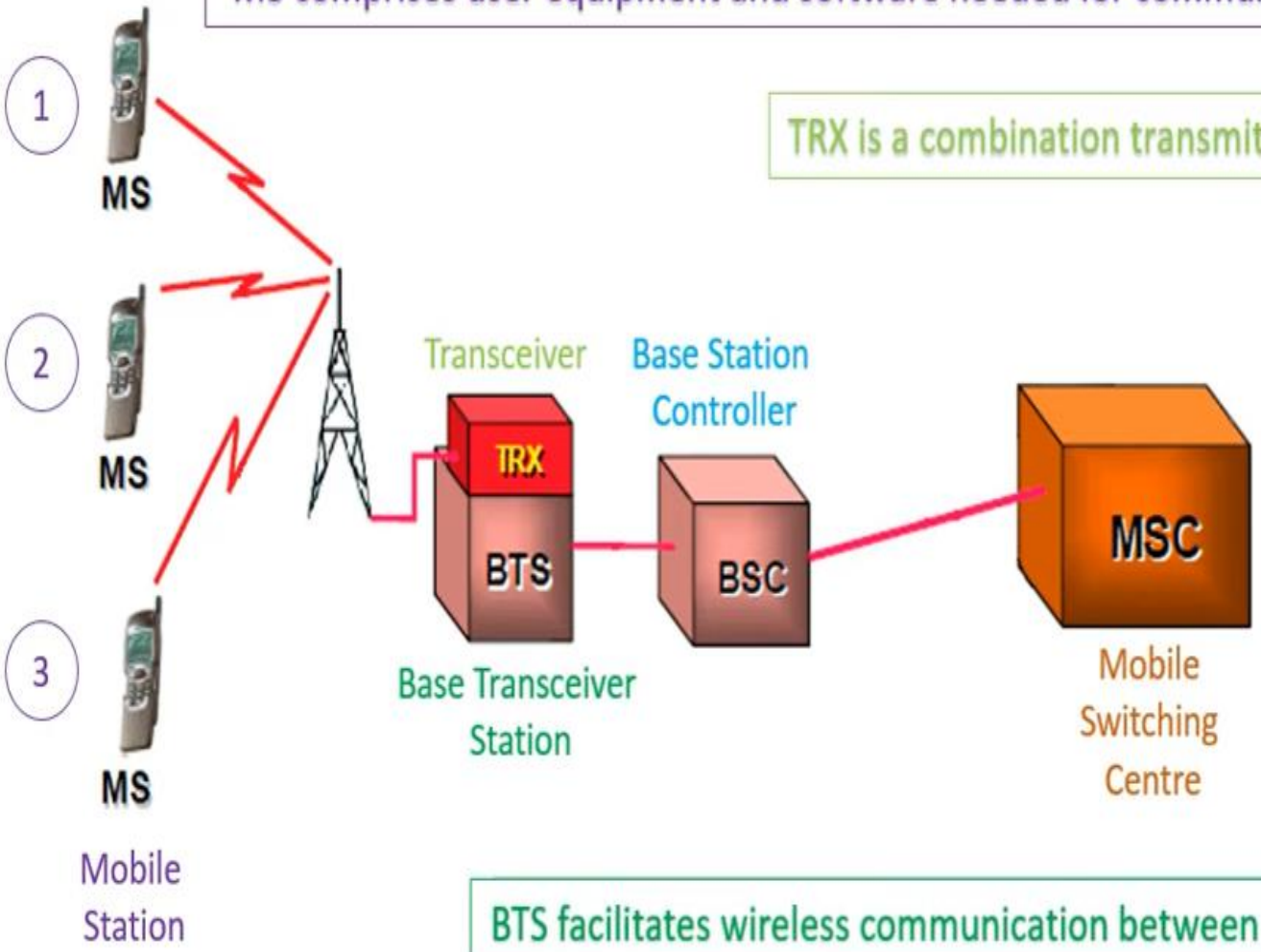
GSM Architecture

The basic structure of cellular system is as below:

- The basic cellular system consists mobile stations, base stations, BSC (Base station controller) and a mobile switching center(MSC).
- The mobile switching center is sometimes called a mobile telephone switching office (MTSO), since it is responsible for connecting all mobiles to the PSTN via central office(CO).
- Each user communicates via radio from a cellular telephone set to the cell site base station.
- This base station connected via telephone lines or microwave link to the mobile switching center.
- The MSC connects the user to the called party if the called party is land based, the connection is via the central office (CO) is the terrestrial telephone network

MS comprises user equipment and software needed for communication with a mobile network

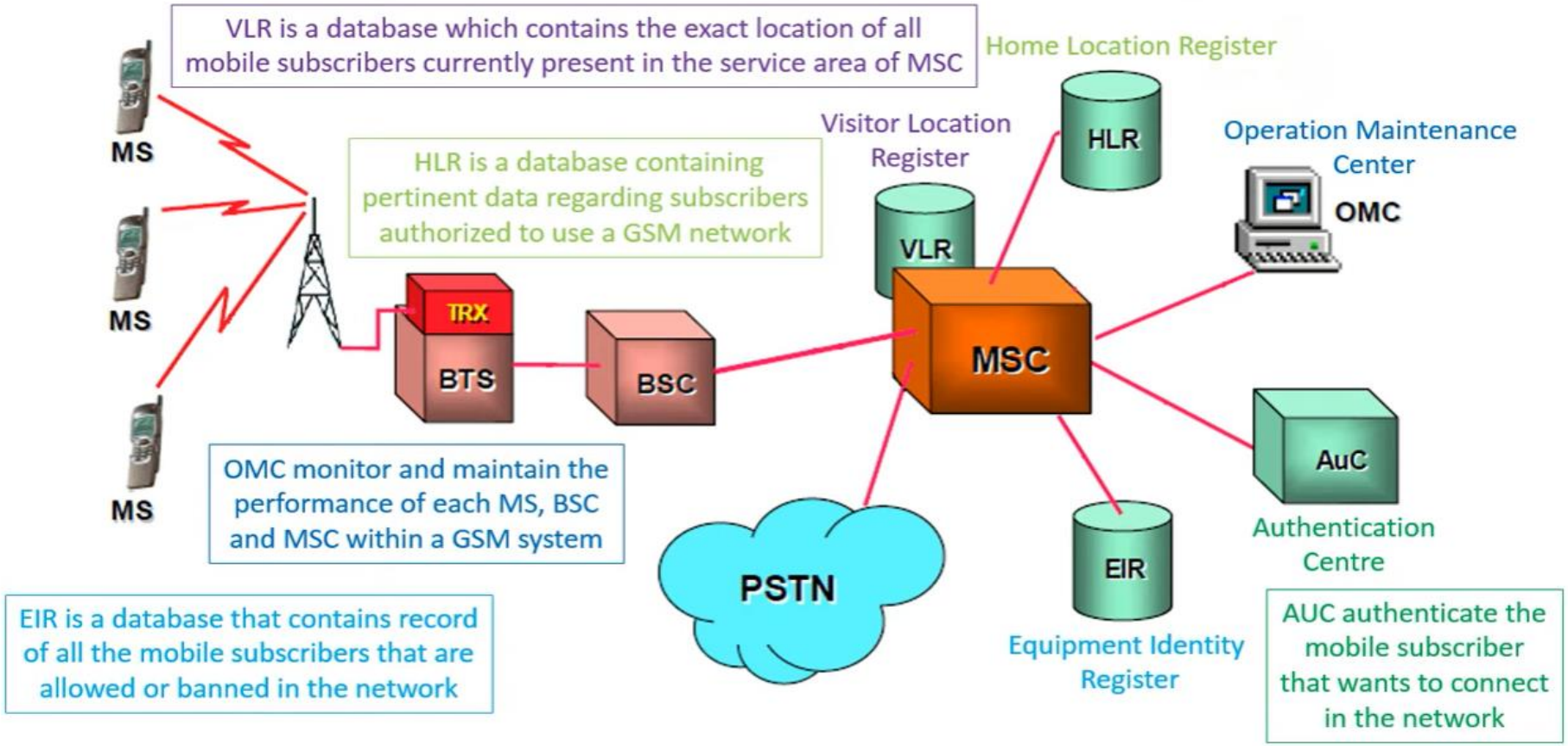
TRX is a combination transmitter/receiver in a single package



Mobile Station (MS)
=
Mobile Equipment (ME)
+
Subscriber Identity Module (SIM)

BTS facilitates wireless communication between user equipment and a network

BSC controls one or more base transceiver stations (BTS)

MSC is associated with communication switching functions such as call set-up, release and routing

For knowledge